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A

FINAL REPORT ON
DESIGN, ANALYSIS AND SIMULATION OF CROSS FLOW TURBINE USING NACA-6409 AIRFOIL AND COMPARISION WITH STANDARD BLADE DESIGN

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ABSTRACT

Cross-flow turbines are widely renowned for their efficiency regardless of changes in water outflow (30%-100%). Despite the fact that this turbine has several benefits, the cross-flow turbine has a considerably lower efficiency than other turbines. Because of eddy formation, the drag force is typically present when water is moving around an immersed body, such as a turbine blade. Typically, this force lowers the turbine's efficiency. It has been demonstrated that airfoil profiles can boost some turbine efficiency by reducing eddy generation in the water flow around immersed bodies, such as turbine blades. The purpose of this study is to determine how the NACA airfoil's blade profile affects cross-flow turbine efficiency. Because it possesses the highest lift coefficient compared to others.

In this work, a cross-flow turbine with NACA-6409 airfoil profiled blades will be compared to a standard turbine using CFD simulation. 20 m head and 0.5 m^3/s of water discharge was assumed. Different design parameters of the standard cross flow turbine was calculated on the basis of the papers published by several authors. 3D model of the turbine was developed using SOLIDWORKS 2022 on the basis of obtained design parameters. The simulation of the turbine was done on ANSYS. From the simulation the efficiency of the standard blade turbine was found to be 71.6%. For airfoil blade the maximum efficiency of 72.3% was obtained.

Keywords: eddy formation, airfoil, NACA-6409, CFD

LIST OF ABBREVIATIONS

AC	Alternating Current
CFD	Computational Fluid Dynamic
CFT	Cross Flow Turbine
FEA	Finite Element Analysis
kw	Kilo Watt
MW	Megawatt
RPM	Revolution Per Minute

LIST OF SYMBOLS

D1	outer diameter of runner
D2	inner diameter of runner
Ds	predicted shaft diameter
Do	pitch circle diameter
R1	runner outer radius
R2	runner inner radius
So	jet thickness
r	curvature of blade
Ro	pitch circle radius
T	blade thickness
N	number of intermediate disc
P	poles of generator
P	power
H	net head
N	speed in RPM
Q	discharge
A	angle of attack
B	blade angle

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CHAPTER 1: INTRODUCTION

1.1 Background

With the continuous depletion of fossil energy reserves, alternative energy as a hydro power plant has becomes a major issue in the provision of energy. The possibility of energy crisis in the future has drawn researcher's attention in founding other sources of green energy these days. Among all the renewable resources, water sources are found abundantly and is relatively a perfect choice and it has a great potential in commercialization and mass production. Now this major application as power generation is becoming increasingly popular.

Hydropower has been a building block in power generation for hundreds of years. Humans have been harnessing the energy of river currents for centuries, using water wheels spun by rivers initially to process grains and cloth. Demands of the clean fuels that are more efficient with natural resources are increasing in recent times. Hydropower is a clean fuel source. It means that it won't pollute the air like power plants that burn fossil fuels such as coal or natural gas [1]. With zero greenhouse emission, zero input cost, low cost per unit energy and sustainability, power generation using hydro resources is preferred.

1.2 Hydropower energy and it's scenario in Nepal

The theoretical potential of Nepal is estimated to be around 83000 MW and the techno-feasible potential is 45610MW [2]. But, only 2% of the potential is harnessed so far which leads to long lasting electricity crisis.

The total electricity made available by NEA, Independent Power Producers (IPPs) and imports is 7741GWh. Out of this, NEA generates only 39.02% on its own. Independent Power Producers (IPPs) supplies 38.64% of the demand and the rest i.e., 22.33% is imported from our neighboring country, India. Similarly, the consumer's growth rate is 8% for the year 2020[3]. It clearly illustrates the need for more production of hydropower energy.

Most of the hydropower installed in Nepal are of conventional type. The hilly and the mountainous region of Nepal have streams and rivers with high head, so most of the research and study have been limited to the development of efficient high head turbines in these regions. There hasn't been enough study and research work for the development of efficient Pico hydro turbines to harness water from location with low head and high discharge.

Table 1.1 Hydropower Categorization in Nepal [4]

S.N.	Category	Capacity
1.	Pico Hydro	< 5 KW
2.	Micro Hydro	5KW to 100 KW
3.	Mini Hydro	>100 KW to 1 MW
4.	Small Hydro	> 1 MW to 10 MW
5.	Large Hydro	> 10 MW

1.3 Micro hydropower in Nepal

Micro-hydro refers to the small-scale hydropower whose power potential ranges between 5KW to 100KW. Micro hydro power plants (MHPs) are a promising electrification technology that uses small water rivers and reservoirs for generating electricity. These kinds of hydropower seem economical for rural electrification.

The use of micro hydro technology has been a success story for Nepal for the several past years. Micro hydro technology has been providing energy to the people living in different communities in 55 districts out of 77 districts [6]. So far, 3300 MHPs with capacity up to 100KW have been installed so far. Around 350000 households have been benefitted with electricity at least for light [5]. There are many types of Micro hydro turbines that can be used to harness hydropower.

1.4 Pico Hydropower

Pico-hydro is a term for very small hydropower units, smaller than „micro-hydro" in the 100W to 5kW range. There are estimated to be over 25,000 water mill sites across Nepal.

It is thought that at least 20% of these could be possible sites for low-head Pico-hydro units. As Compact Fluorescent Light (CFL) technology has become cheaper and more affordable for villagers, a smaller amount of power generated can go a lot further in terms of lighting more households. 1 kW of power can provide over 100 CFL bulbs (9W) for a community. So, Pico hydro can be very beneficial smaller villages that cannot afford a larger hydropower project [7].

1.5 Hydraulic Turbine

A hydraulic turbine is a prime mover (a machine which uses the raw energy of a substance and converts it into mechanical energy) that uses the energy of flowing water and converts it into

the mechanical energy (in the form of rotation of the runner). This mechanical energy is used in running an electric generator which is directly coupled to the shaft of the hydraulic turbine; from this electric generator, we get electric power which can be transmitted over long distances by means of transmission lines and transmission towers. The hydraulic turbines are also known as 'water turbines' since the fluid medium used in them is water [8].

The different types of hydraulic turbines are mentioned below:

1.5.1 Pelton

A Pelton turbine or Pelton wheel is a type of hydro turbine (specifically an impulse turbine) used frequently in hydroelectric plants. These turbines are generally used for sites with heads with very high heads and low discharge.

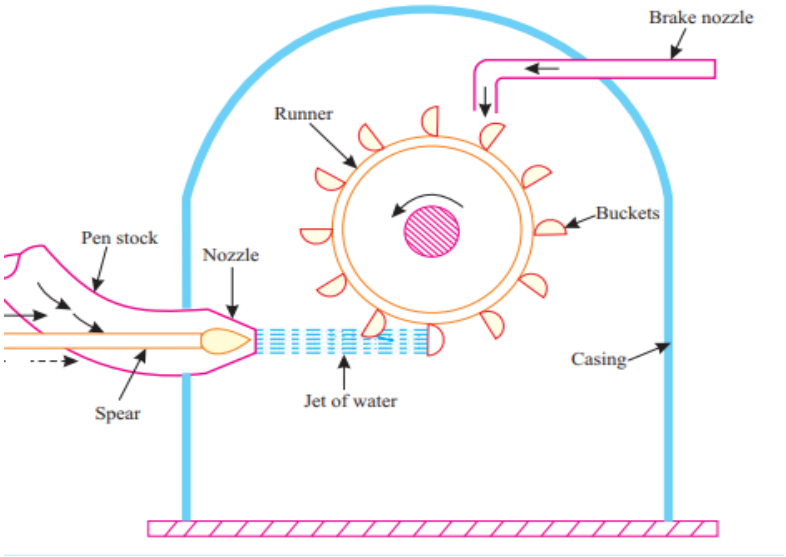


Figure 1.1 Pelton turbine

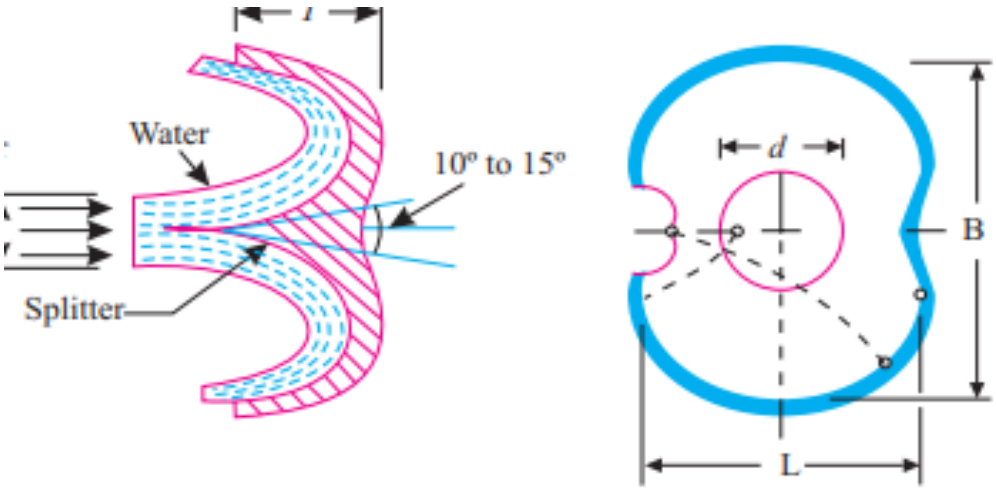


Figure 1.2 Front view and side view

1.5.2 Francis

The modern Francis turbine is an inward mixed flow reaction turbine (in the earlier stages of development, Francis turbine had a purely radial flow runner), i.e. water under pressure, enters the runner from the guide vanes towards the center in radial direction and discharges out of the runner axially. The Francis turbine operates under medium heads and also requires medium quantity of water. It is employed in the medium head power plants. This type of turbine covers a wide range of heads. Water is brought down to the turbine through a penstock and directed to a number of stationary orifices fixed all around the circumference of the runner. These stationary orifices are commonly called as guide vanes or wicket gates

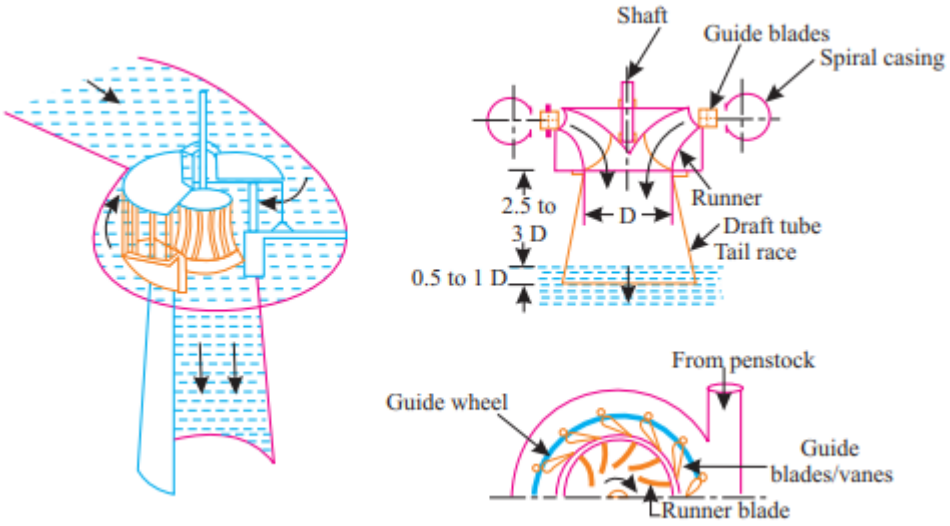


Figure 1. 3 Francis Turbine

1.5.3 Kaplan

A propeller turbine is quite suitable when the load on the turbine remains constant. At part load its efficiency is very low; since the blades are fixed, the water enters with shock (at part load) and eddies are formed which reduce the efficiency. This defect of the propeller turbine is removed in Kaplan turbine. In a Kaplan turbine the runner blades are adjustable and can be rotated about pivots fixed to the boss of the runner. The blades are adjusted automatically by

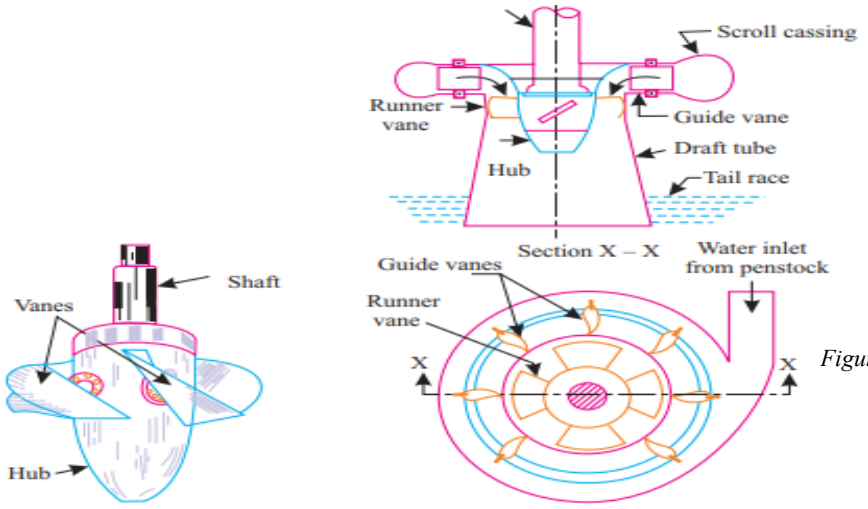


Figure1.4 Kaplan turbine

servomechanism so that at all loads the flow enters them without shock. Thus, a high efficiency is maintained even at part load. The servomotor cylinder is usually accommodated in the hub.

1.5.4 Cross flow Turbine (Banki)

Cross-flow turbine, Bánki-Michell turbine, or Ossberger turbine is a water turbine developed by the Australian Anthony Michell, the Hungarian Donát Bánki and the German Fritz Ossberger. Michell obtained patents for his turbine design in 1903. Unlike most water turbines, which have axial or radial flows, in a cross-flow turbine the water passes through the turbine transversely, or across the turbine blades. As with a water wheel, the water is admitted at the turbine's edge. After passing to the inside of the runner, it leaves on the opposite side, going

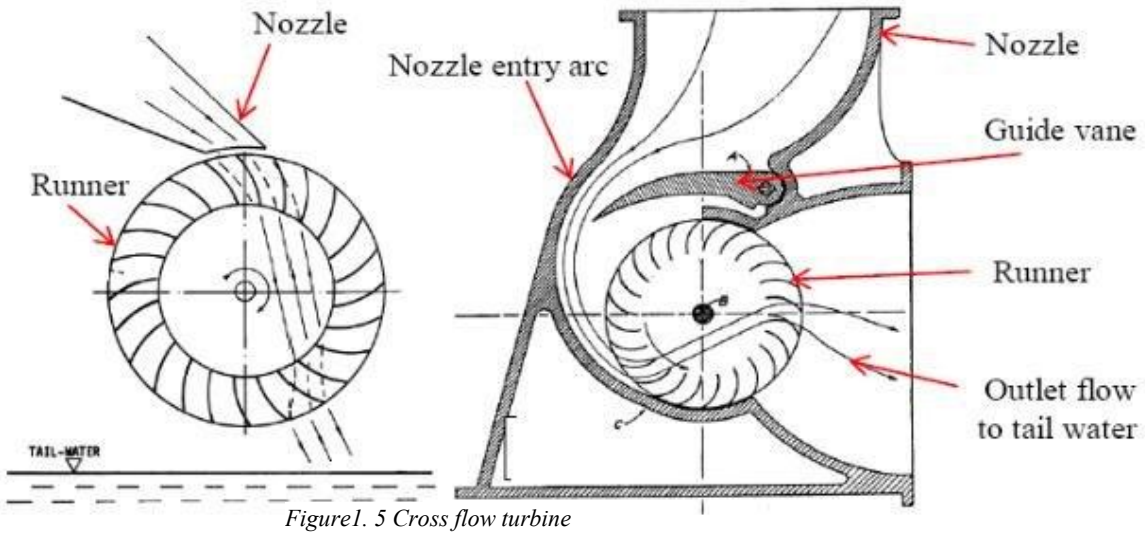


Figure1. 5 Cross flow turbine

outward. Passing through the runner twice provides additional efficiency. When the water leaves the runner, it also helps clean it of small debris and pollution. The cross-flow turbine is a low-speed machine that is well suited for locations with a low head but high flow. Cross-flow turbines are often constructed as two turbines of different capacity that share the same shaft. The turbine wheels are the same diameter, but different lengths to handle different volumes at the same pressure. The subdivided wheels are usually built with volumes in ratios of 1:2. The subdivided regulating unit, the guide vane system in the turbine's upstream section, provides flexible operation, with 33, 66 or 100% output, depending on the flow. Low operating costs are obtained with the turbine's relatively simple construction.

The turbine consists of a cylindrical water wheel or runner with a horizontal shaft, composed of numerous blades, arranged radially and tangentially. The blade's edges are sharpened to reduce resistance to the flow of water. A blade is made in a part-circular cross-section. The ends of the blades are welded to disks to form a cage like a hamster cage and are sometimes

called "squirrel cage turbines"; instead of the bars, the turbine has the trough-shaped steel blades.

The water flows first from the outside of the turbine to its inside. The regulating unit, shaped like a vane or tongue, varies the cross-section of the flow. The water jet is directed towards the cylindrical runner by nozzle. The water enters the runner at an angle, transmitting some of the water's kinetic energy to the active cylindrical blades.

The regulating device controls the flow based on the power needed, and the available water. The turbine geometry (nozzle-runner-shaft) assures that the water jet is effective. The water acts on the runner twice, but most of the power is transferred on the first pass, when the water enters the runner. Only ⅓ of the power is transferred to the runner when the water is leaving the turbine. The water flows through the blade channels in two directions: outside to inside, and inside to outside. Most turbines are run with two jets, arranged so two water jets in the runner will not affect each other. It is, however, essential that the turbine, head and turbine speed are harmonized. The cross-flow turbine is of the impulse type, so the pressure remains constant at the runner.

1.6 History of Cross Flow Turbines in Nepal

In 1960 the Swiss technical co-operation agency Helvetas established the first mechanical workshop in Nepal, to create a minimal industrial infrastructure needed to execute repair works for equipment used in its other projects. At that time ther was no any skilled manpower was available. So a mechanical training centre was incorporated in 1964. Fast forward in time it became among the biggest enterprises in Nepal, named BYS (Balaju Yantra Shala P (Ltd.)).

In the late sixties the company started to develop a simple water turbine. SKAT, the Swiss Centre for Appropriate Technology is collaborating with BYS mainly in the field of micro hydro power (MHP) technology. It was observed that locally produced water mills were properly working but were not powerful enough to drive rice husking machinery. A large number of Nepalese farmers knew that in India diesel engines were available to drive small flour mills and rice huskers. Upon farmer’s pressure, the Nepalese Agricultural Development Bank (ADB-N) started granting loans for diesel mills. At this time BYS came up with the idea to produce turbine driven mills. A Swiss turbine manufacturer was approached to furnish BYS with the technical drawings for a propeller turbine. There was no foundry in Nepal, therefore the design had to be adapted but by people never having designed any turbine. The blade profiles of the runner could not be produced with the machinery in BYS. However, a few

prototypes at probably half the efficiency of the original design went into operator and marked a breakthrough in the sense, that it became public that rice hullers could be driven with hydropower. With this break through the BYS was the first to introduce cross flow turbine in Nepal [9].

1.7 Airfoil

An airfoil (or aerofoil in British English) is any structure designed to manipulate the flow of a fluid to produce a reaction, which in an aircraft's case, is aerodynamic lift. The wings of fixed-wing aircraft feature airfoil-shaped cross-sections. Airfoils enable heavier-than-air flight, but are also in found various other vehicle parts like helicopters, turbines, and spoilers.

In this project we have selected an airfoil of NACA-6409. It is an airfoil having the maximum mean camber of 6% at the position of 40% of cord from the leading edge. The thickness of the airfoil is 09%.

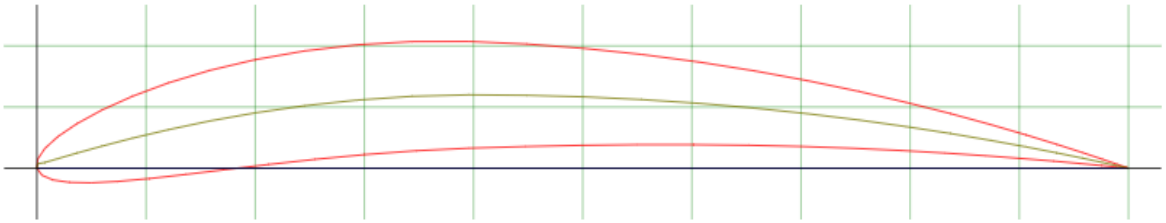


Figure1. 6 NACA-6409

(Source: http://airfoiltools.com/)

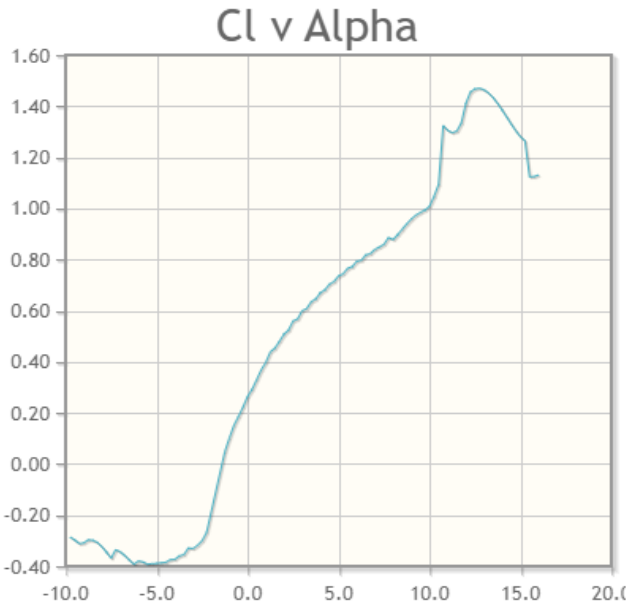


Figure1.8 cl vs alpha

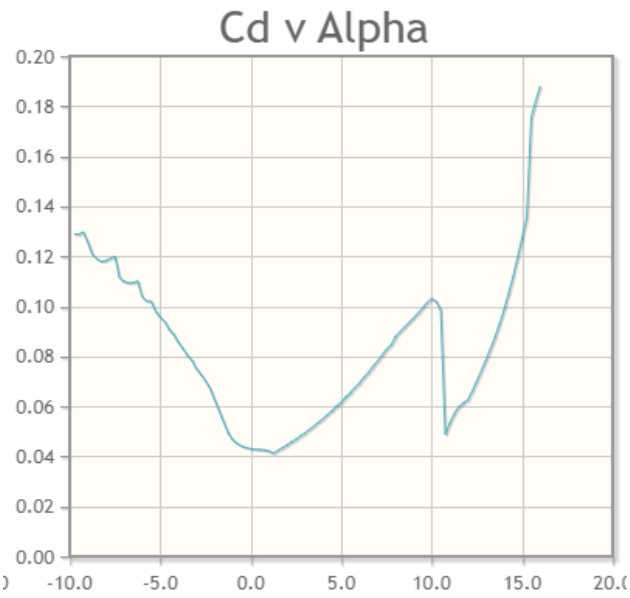


Figure1.7 cd vs alpha

1.7 Problem Statement

While the whole world is being electrified, many rural areas are not connected to the central electric grid. The micro hydropower gives a promising hope in the rural electrification. mainly cross flow type of turbines are used to harness the power from small flow of water. In spite of the simple construction and light weight. Its efficiency in the large range of 60-70%. The large hydropower plants can harness about 90 to 95% of the the available power. Our study aims at improving the cross-flow turbine efficiency by changing the cross section of the blade of the turbine runner.

1.8 Objective

Main objective

- To compare the efficiency between standard turbine and airfoil shaped blade turbine

Specific objective

- To design the CFT by calculating the different parameters
- To make the 3D model from the calculated parameter
- To simulate and obtain efficiency
- Compare the efficiencies

1.9 Scope / Significance

This project specially focuses on designing CFT by using airfoil shape blade instead of curved shaped blade and simulate it and observed the simulation to increase the efficiency. By using airfoil shape we may obtain increase in power output and efficiency.

CHAPTER 2: LITETRATURE REVIEW

2.1 Related Works

Cross-flow turbine is composed of two major parts, the runner and the nozzle. The runner is a circular rotor with two side walls to which the blades are fixed along the periphery of the turbine and blades are circular with specific radius of curvature. The nozzle directs the water flow into the runner at a certain attack angle. The water flow comes through an inlet pipe, and then it is regulated by guide vanes and finally enters the runner of the turbine. After passing through, the water leaves on the opposite side of the runner, providing so additional efficiency. The efficiency of the cross-flow turbine is dependent on several design parameters. These include runner diameter, runner length, runner speed, turbine power, water jet thickness, blade spacing, number of blades, radius of blade curvature, attack angle and the blade and exit angles[9].

M. Sinagra et.al ," Cross-Flow turbine design for variable operating conditions" (2014)

Cross-flow or Banki-Michel turbines are a very efficient and economic choice that allows a very good cost/benefit ratio for energy production located at the end of conduits carrying water from a water source to a tank. In the paper the optimum design of a cross-flow turbine was sought after, assuming a flow rate variable in time.

Regulation of the discharge entering in the turbine was a key issue, which was faced adopting a shaped semicircular segment, moved inside the main case around the rotating impeller. The maximum efficiency of the turbine was attained by setting the velocity of the particles entering the impeller at about twice the velocity of the rotating system at the impeller inlet. Energy losses along the pipe were considered negligible, closing and opening the inlet surface with the semicircular segment allows always a constant hydraulic head and a constant velocity at the impeller inlet, even with different flow rate entering values. Observed reduction of the turbine efficiency along with the inlet surface reduction was first investigated; a design methodology, using also CFD simulations, was then proposed. The paper presented a simple and rigorous procedure for the design of a cross flow turbine with discharge regulator which allows a constant upstream hydraulic head.

C.H. Achebe, et.al." Design and implementation of a crossflow turbine for Pico hydropower electricity generation" (2020)

The study covered the design, implementation and performance evaluation of a crossflow turbine at various nozzle positions. The chosen blade material was analyzed using ANSYS for stress and deformation degree under the impact of hydraulic jets to ascertain its suitability while in operation. The shaft was analyzed under static and dynamic conditions using ANSYS in order to ensure a non-plastic deformation of the shaft. A crossflow turbine was constructed using computed design values for all the machine's parts. The runner blades were positioned

specifically at 28^0 outer blade angle and 90^0 inner blade angle. The turbine was tested under a water head and flow rate of 6.4m and $0.0042m^3/s$ respectively. The shaft power and efficiency were evaluated using their respective formula. The best turbine performance was obtained at lower angle of attack, nozzle distance very close to the runner shaft and at a nozzle height that will actualize greater energy impartation to the upper and lower blade profiles. An optimal nozzle distance, height and attack angle of 102mm, 413mm and 5 respectively, were obtained. At this nozzle position, the alternator gave an output of 35watts and 6V.

Manchan Tiwari , Rajendra Shrestha," Effect of Variation of Design Parameters on Cross Flow Turbine Efficiency Using ANSYS"(2017)

In this research, Cross flow turbine designs were obtained from local manufacturers. Efficiencies of these turbines were determined using simulation under steady state condition. Efficiencies were verified using the data from the installation site where these designs were used. Different Cross flow turbine models were prepared by varying the curvature radius of the blade and the ratio of inner to outer radius of the runner. The efficiencies of such models were determined using simulation. Among the various Cross flow turbine models which were prepared by varying the design parameters, most efficient model was found to be with the radius of blade curvature 0.045 m and diameter ratio 0.7. As the value of the varied design parameters were taken to the close proximity of the standard design values, the results showed room for improvement of the efficiency by taking wide range of variation of the radius of blade curvature and the diameter ratio.

CARIJA, Z, et.al," FLUID FLOW SIMULATION OF A CROSSFLOW TURBINE(2009)

In the paper the numerical 2D multiphase fluid flow simulation through a crossflow turbine in a specified operating point were done. The results showed expected turbine efficiency and a realistic flow distribution. This provided a basis for further simulations aimed at improving the turbine design. A Solid model of the turbine was created in CATIA according to available designs. The runner had 24 blades, an outer diameter of 300 [mm] and a rotational speed of 428.6 $[min^{-1}]$. The analysis showed the computed turbine efficiency to be $\eta = 59.8\%$ and the power obtained was P = 24.65 [kW]. The first pass through the runner allows the largest part of the kinetic energy of water (around 70%) to be converted into useful kinetic energy of rotation, while the remaining (30%) harnessed energy is converted when the water exits the runner into the draft tube.

Dendy Adanta, et.al," Effect of Blade Depth on the Energy Conversion Process in Crossflow Turbines"(2020)

This study discussed the effect of blade depth on the energy conversion process in crossflow turbines. Using computational methods, this study compares three blade depths: 1.5-mm, 3-mm and 4.5-mm. Two-dimensional transient simulations were carried out using the six degrees of freedom (6 DoF) approach. The viscous model was shear stress transport (SST) k-ω and 6 with the volume of fluid (VoF) approach. From the results, the maximum efficiency of crossflow turbines shows that blades with greater depths tend to have higher efficiency.

However, the 3-mm blade depth showed maximum efficiency vulnerable to depths wider than 4.5 mm and 1.5 mm.

Young-Do Choi,et.al," Effect of Blade Angle on the Performance of a Cross-Flow Hydro Turbine"(2008)

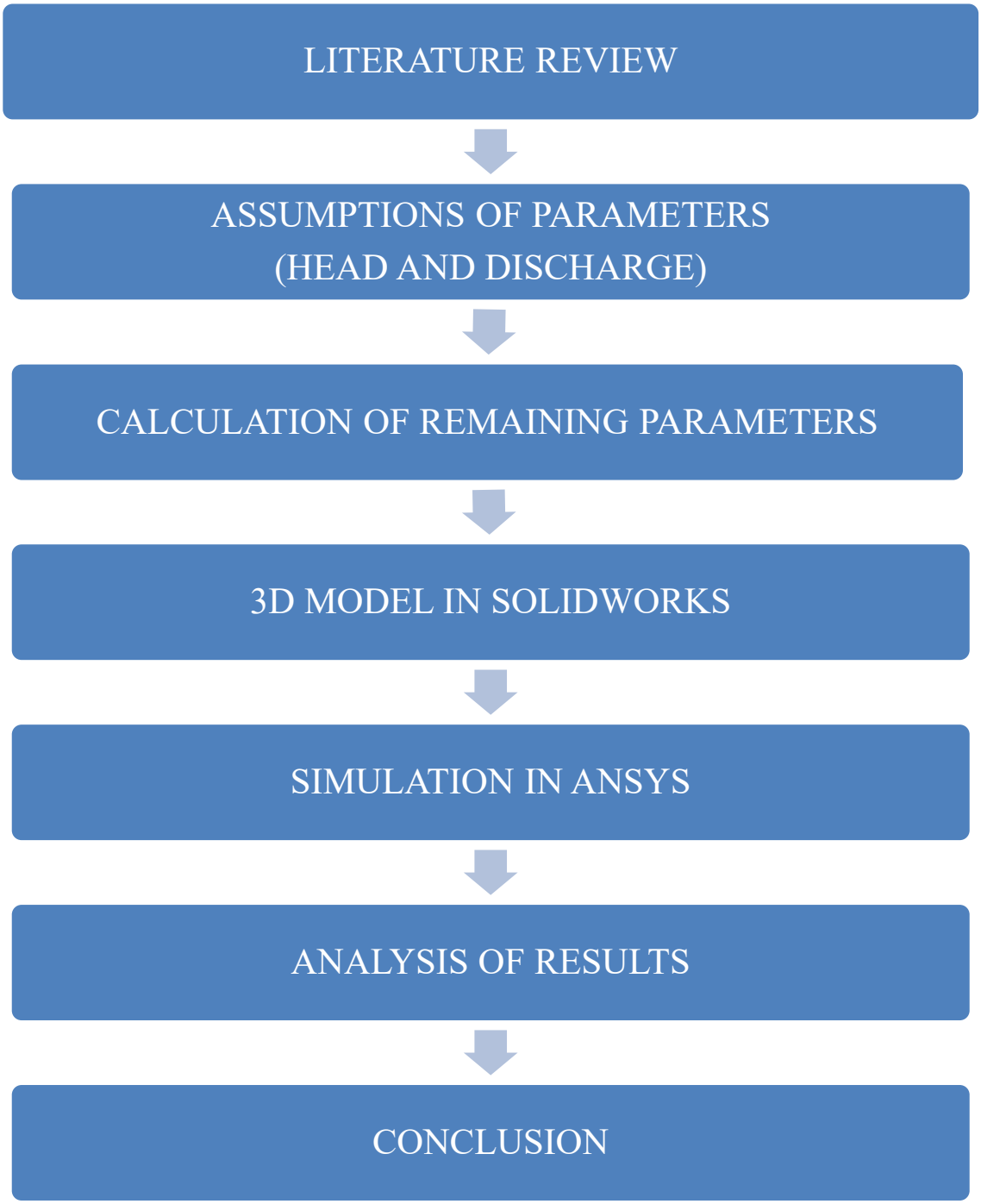
This study aimed to investigate the effect of blade angle on the performance of the cross-flow hydro turbine. Analysis of the turbine performance with the variation of the blade angle has been made by using a commercial CFD code. The results show that inlet and outlet angles of runner blade give considerable effect on the performance of the turbine. Pressure on the surface of the runner blade changes remarkably by the blade angle both at the Stages 1 and 2. Moreover, relatively small blade inlet angle is effective to produce higher value of output power. Recirculating flow in the runner passage causes remarkable hydraulic loss.

Myin San, Nyi Nyi," Design of Cross Flow Turbine and Analysis of Runner's Dimensions on Various Head and Flow Rate"(2018)

This paper represents the runner design of Cross-flow turbine that will generate 100 kW output power from head of 28 m and flow rate of 0.5 m3 /s. For these head and capacity, rotational speed is 600 rpm, specific speed is 95.39, runner diameter is 340 mm and runner width is 416 mm. In this paper, detail design drawing of runner are also clearly presented. Moreover, the runner's diameter and breadth are represented with various flow rate range of 0.2 to 1 m3 /s and head range of 10 m to 50 m. Thus, it is easy to select the suitable runner's dimensions on the corresponding head and flow rate.

CHAPTER 3: METHODOLOGY

3.1 Workflow Diagram



3.2 Design procedure

To calculate the different parameters of the turbine we first assumed the discharge and head. The discharge is assumed to be 0.5m^3/s and head is assumed to be 20 m. Assuming the overall efficiency the expected power is 65 kw

From the chart below the cross flow turbine is selected.

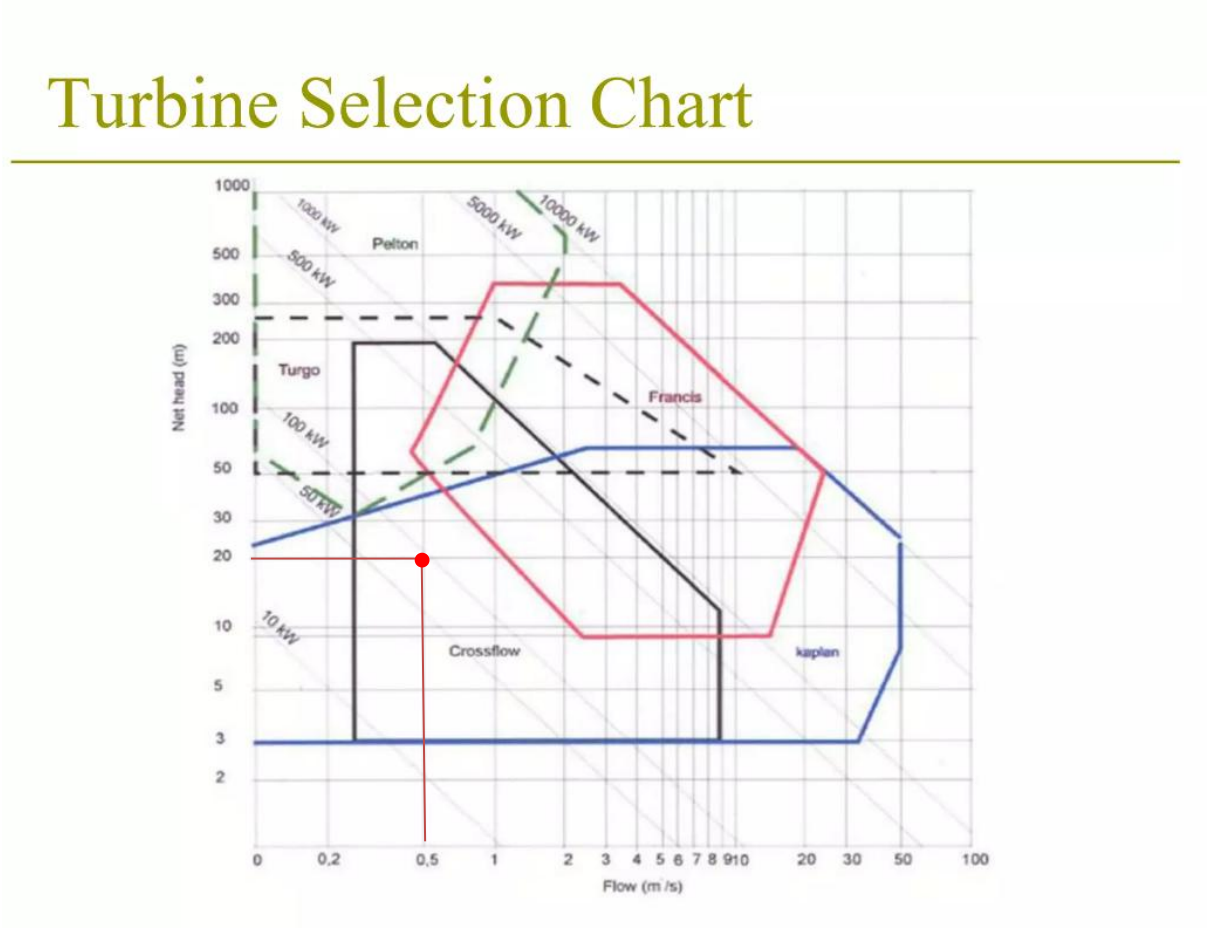


Figure 3.1 Alstom chart

To use 10 pole generator, we assume the RPM of the turbine to be 600 which can be find from the following formula:

$$N = \frac{120 * f}{p} \quad (1)$$

Where,

N= speed in RPM

F= frequency of ac current (500)

p= no. poles of generator

3.2.1. Runner wheel design

The runner of the crossflow turbine consists of the blades positioned at specific angles and the shaft for power transmission to the alternator unit. The ratio of the inner to the outer diameters of the wheel is an imperative parameter of the crossflow turbine design

3.2.2. The outer diameter of the runner wheel (D1)

Mockmore and Merryfield [2] derived an equation relating the outer diameter, flow head and the revolutional speed of turbine as:

$$D1=40\frac{\sqrt{H}}{N} \quad (2)$$

Where,

H= net head

N= speed

3.2.3. Diameter ratio

It is the ratio of inner diameter of runner to the outer diameter of runner. For the standard turbine the diameter ratio is 0.67.

$$\frac{D2}{D1}=0.67 \quad (3)$$

Here,

D2 = runner inner diameter

3.2.4. Thickness of Jet

The thickness of jet varies from 0.1 to 0.2 runner outer diameter. By taking the average nearly.

$$So = 0.17D1 \quad (4)$$

Where

So= jet thickness

3.2.5. Length of Runner

The length of runner can be found by the following formula:

$$L = \frac{0.23Q}{So\sqrt{H}} \quad (5)$$

Where,

L= length of runner

Q= discharge

3.2.6. Blade Design

The outer and inner blade angles, blade thickness, radius of the blade, number of blades, pitch circle diameter, blade spacing, inlet discharge angle and central angle of blade are the blade

design parameters to be considered. The blade curve can be chosen from a circle whose center lies at the intersection of two perpendiculars, one to the direction of the relative velocity and other to the tangent to the inner periphery intersection. The drawing method of blade curvature is shown in Fig.

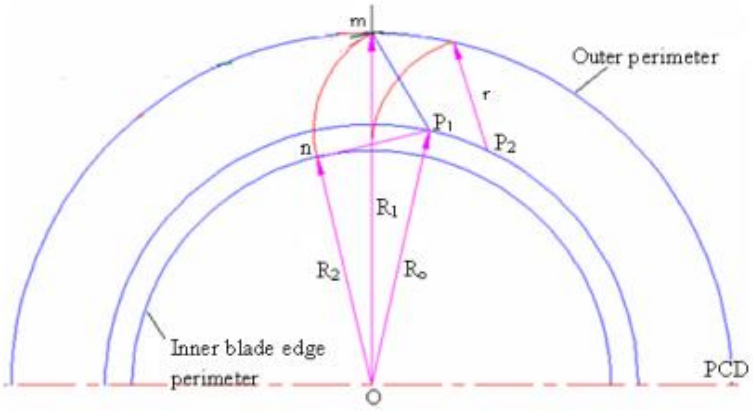


Figure 3.2 blade curvature drawing

Where, Outer radius of runner, R1

Inner radius of runner, R2

Radius of blade shaped arc, r = 0.16 D1

Radius of circle of center pitch of shaped arc, Ro

$$Ro^2=R2^2+r^2 \quad (6)$$

The number of blades, Z is ($24 \leq Z \leq 30$) and the fewer blades may cause pulsating power, while a larger number of blades may cause excessive friction losses. So, the number of blades is chosen as 25 blades in this design.

3.2.7. Blade Thickness

The thickness of the blade depends on the diameter of runner and the relationship between them is shown in Table .(Myin San , Nyi Nyi 2018)

Table I. Runner Diameter and Blade Thickness [2]

Runner Diameter (mm)	200	300	450	700	1000
Thickness of Blade (mm)	3.2	4.5 ~ 6	6	9	12

3.2.8. Prediction of Shaft Diameter

The turbine shaft will transmit the rotary motion of the runner to the generator via the drive system. In most cases, the shaft has a circular cross-section and it subject to either pure torsion or a combination of torsion and bending. The shaft diameter is estimated by

$$d_s=\sqrt[3]{\frac{P}{N}} \quad (7)$$

Where,

d_s=predicted shaft diameter

P=Power developed

N= speed

3.2.9. Angle of Attack and Blade angle

The angle of attack was taken about 16^0 and the blade angle was taken 28^0 .

Mockmore and Merryfield , gave the relationship between the attack angle (α) and outer blade angle (β1) as;

$$\tan\beta 1 = 2\tan\alpha \quad (8)$$

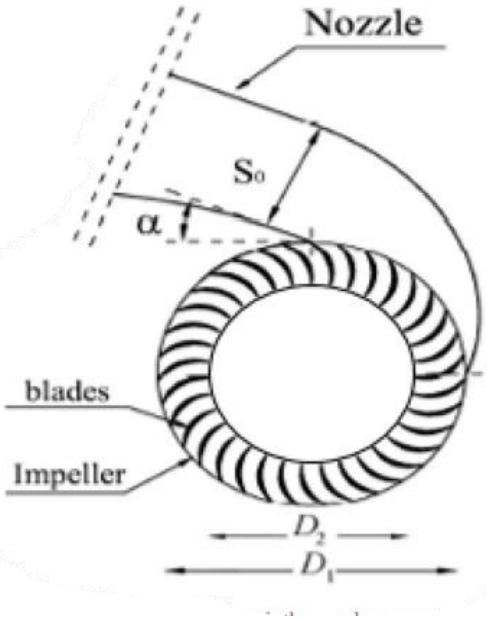


Figure 3.3 runner and nozzle

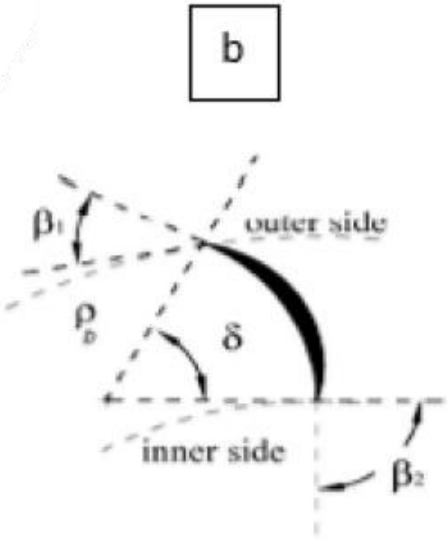


Figure 3.4 blade

3.2.10. Number of Intermediate disc

The number of intermediate disc required in the runner was found from the chart given [10]

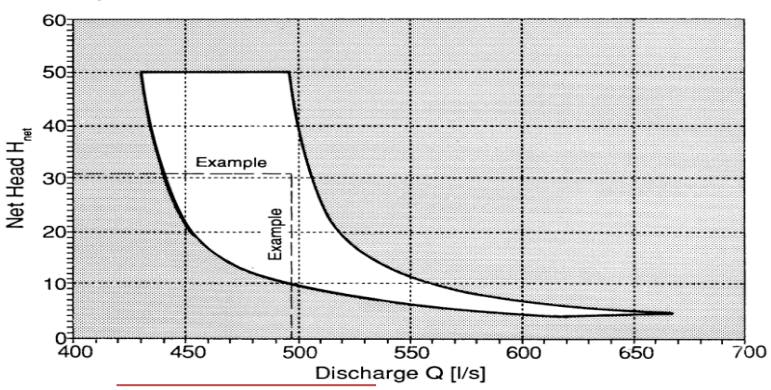


Figure 3.5 selection chart

RESULTS OF RUNNER DESIGN

Table3.1 design parameters

SN	Parameters	Specifications
1	The outer diameter of the runner wheel (D1)	300 mm
2	The inner diameter of the runner wheel (D2)	200 mm
3	Outer blade angle (β1)	30^0
4	Inner blade angle (β2)	90^0
5	Radius of the blade(r)	48 mm
6	Blade thickness (t)	5 mm
7	Pitch circle diameter(D_0)	222 mm
8	Length of the runner(L)	600 mm
9	Number of blades(Z)	25
10	Jet thickness (So)	50 mm
11	Design Volumetric discharge(Q)	500 l/s
12	Number of intermediate discs(n)	3
13	Predicted shaft diameter(ds)	50mm

3.3. 3D CAD Model

The CAD model of the turbine was prepared in SOLIDWORKS using the calculated design parameters.

The different parts of the turbine designed in the solidworks 2022 are:

3.3.1. Casing

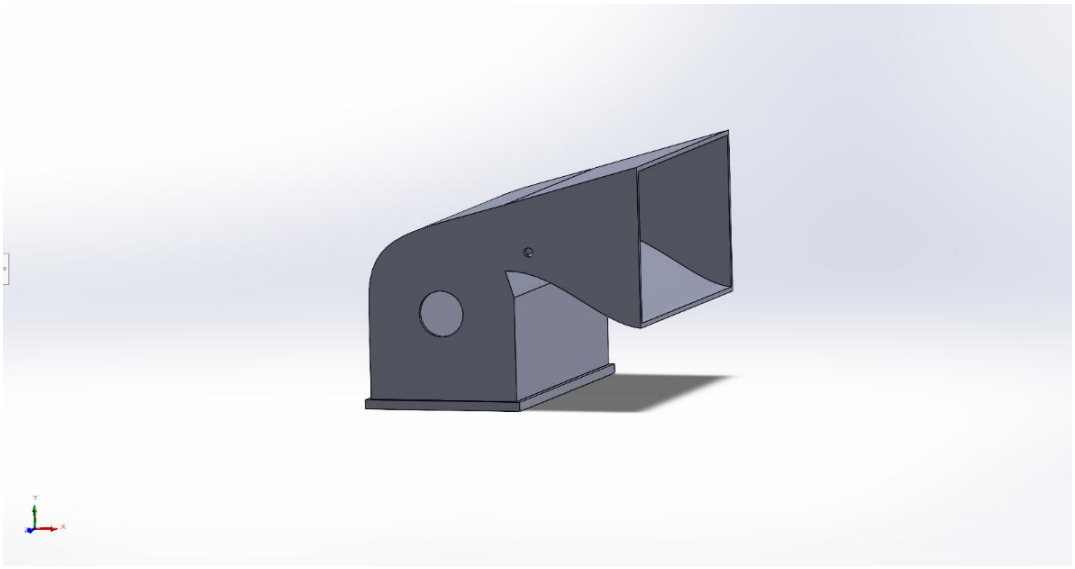


Figure 3.6 casing

3.3.2. Guide Vanes

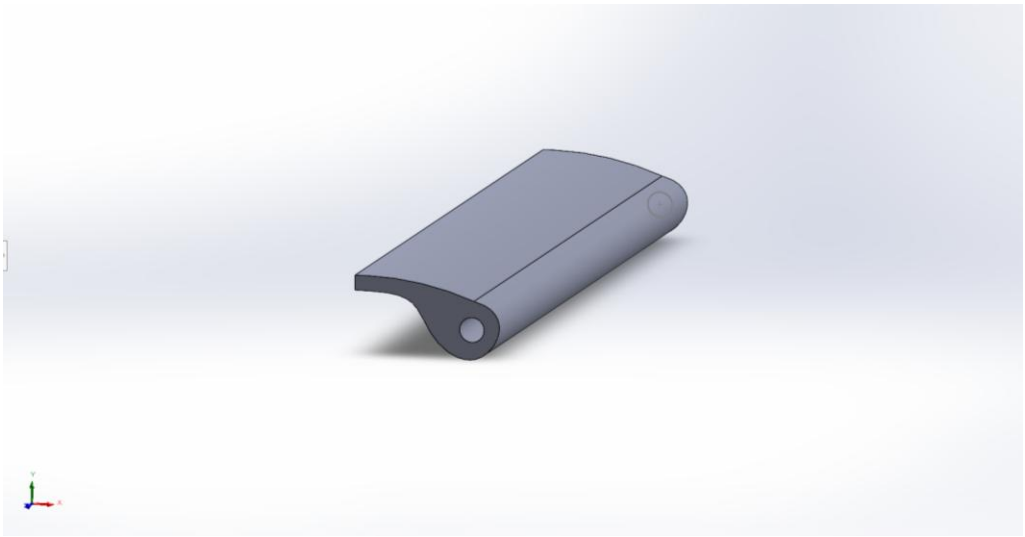


Figure 3.7 guide vanes

3.3.3. Draft Tube

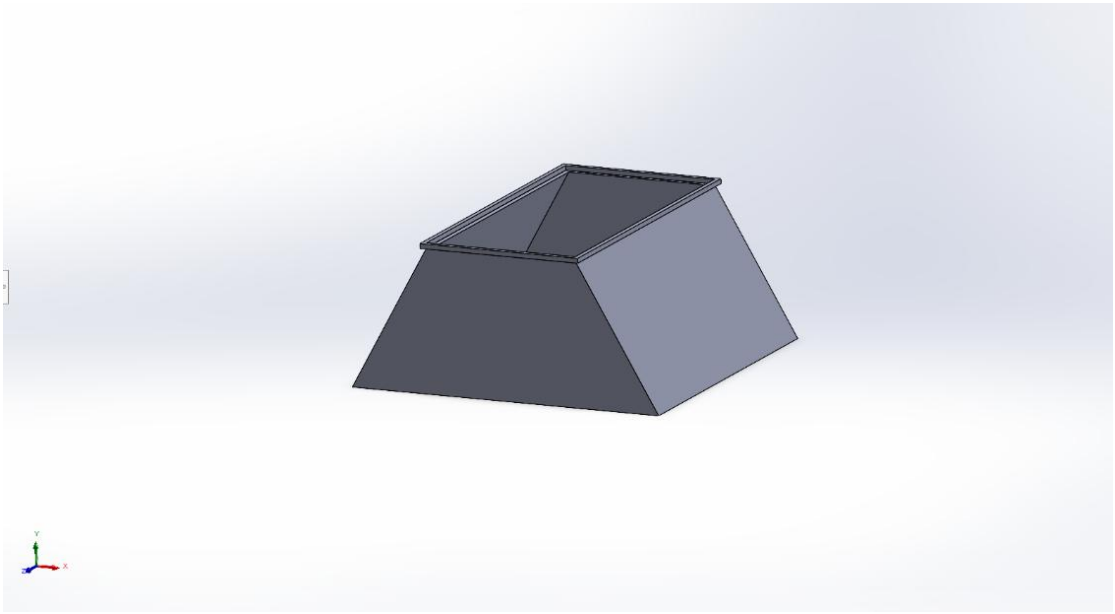


Figure 3.8 draft tube

3.3.4. Standard blade

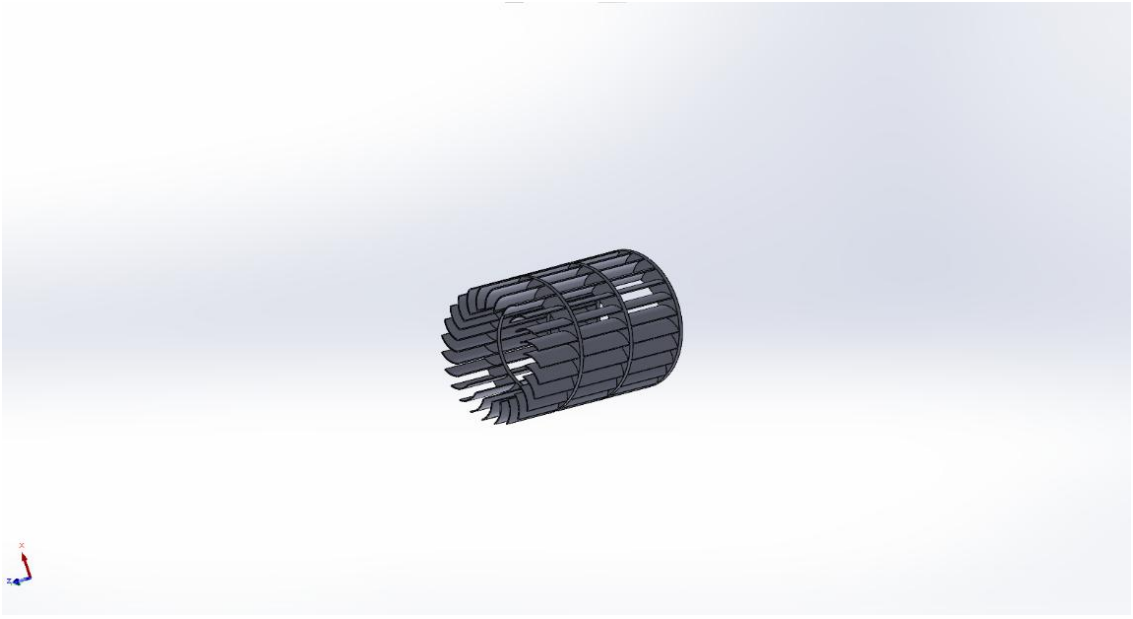


Figure 3.9 section view of standard blade

3.3.5. Airfoil blade

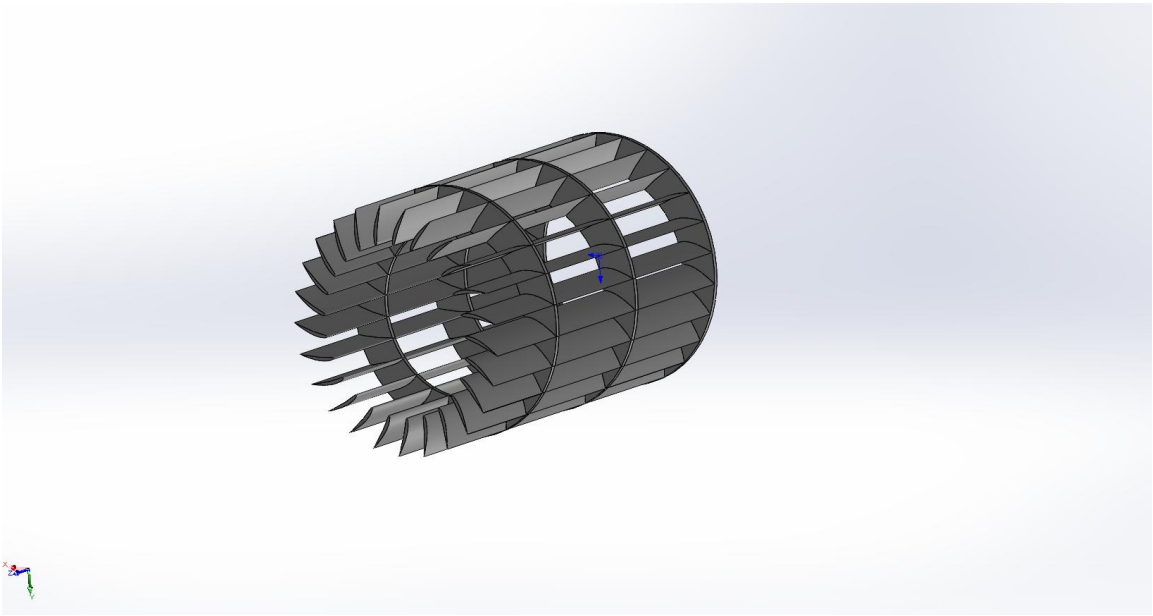


Figure 3.10 section view of airfoil blade

3.3.6. Inverted airfoil blade

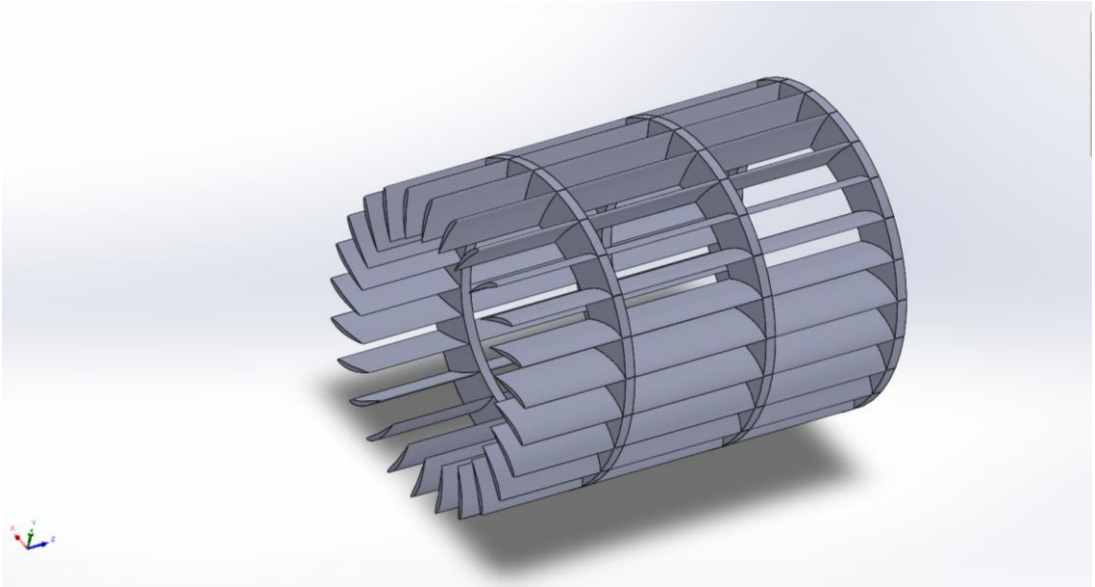


Figure 1.11 section view of inverted airfoil blade

3.3.7. Washer

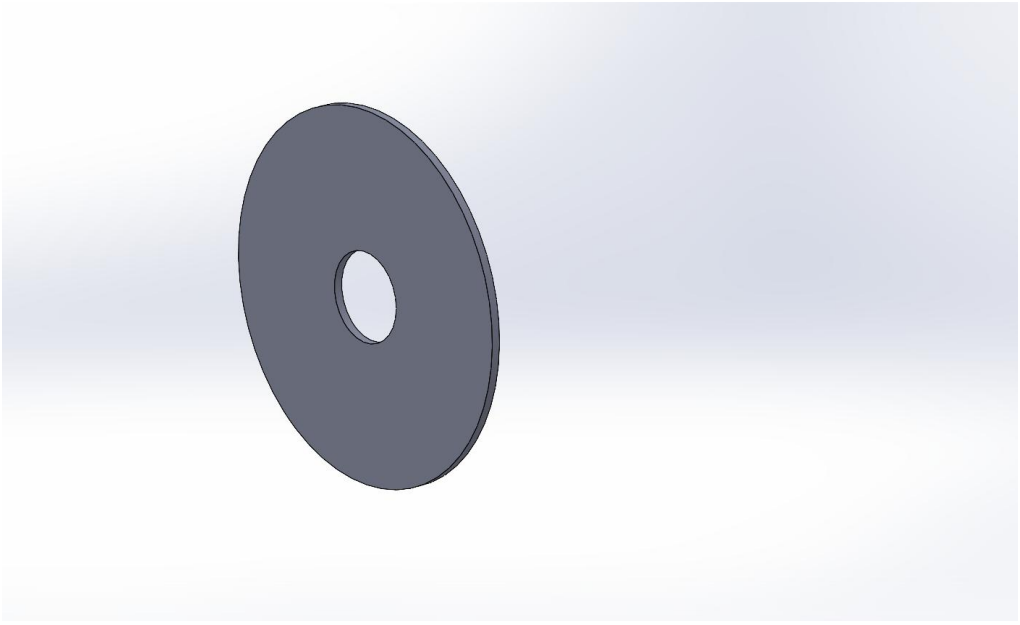


Figure 3.12 washer

3.3.8. Shaft

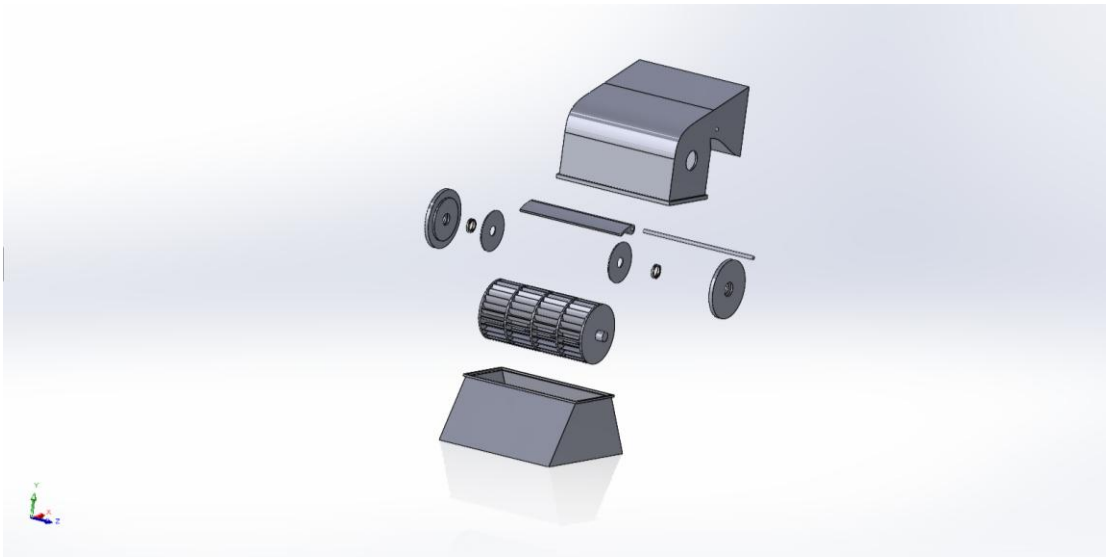


Figure 3.13Exploded View

3.3.9. Bush

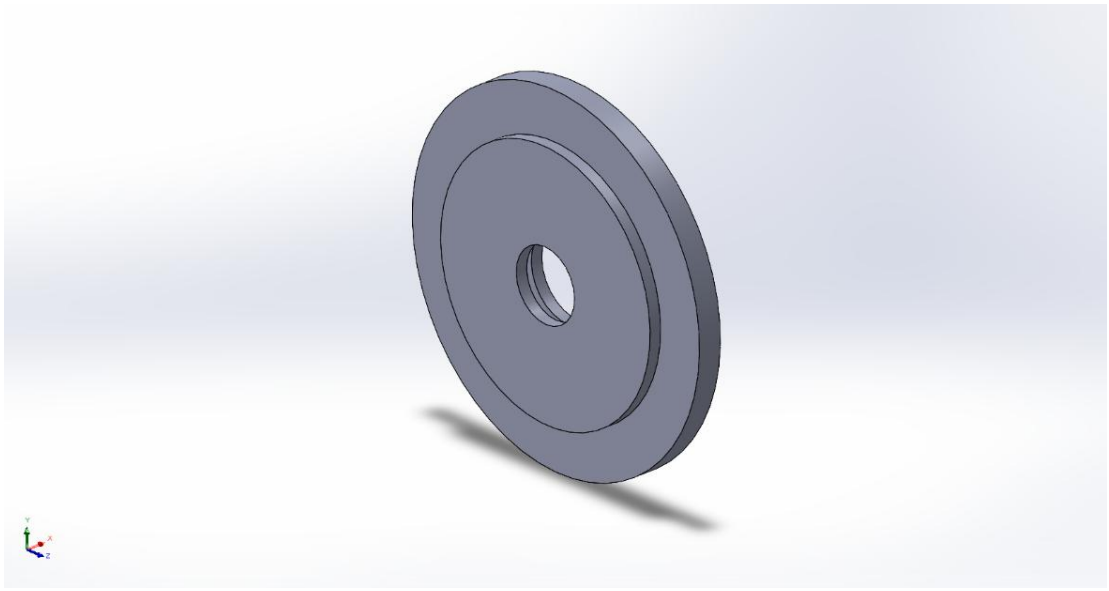


Figure 3.14 bush

3.3.10. Bearing

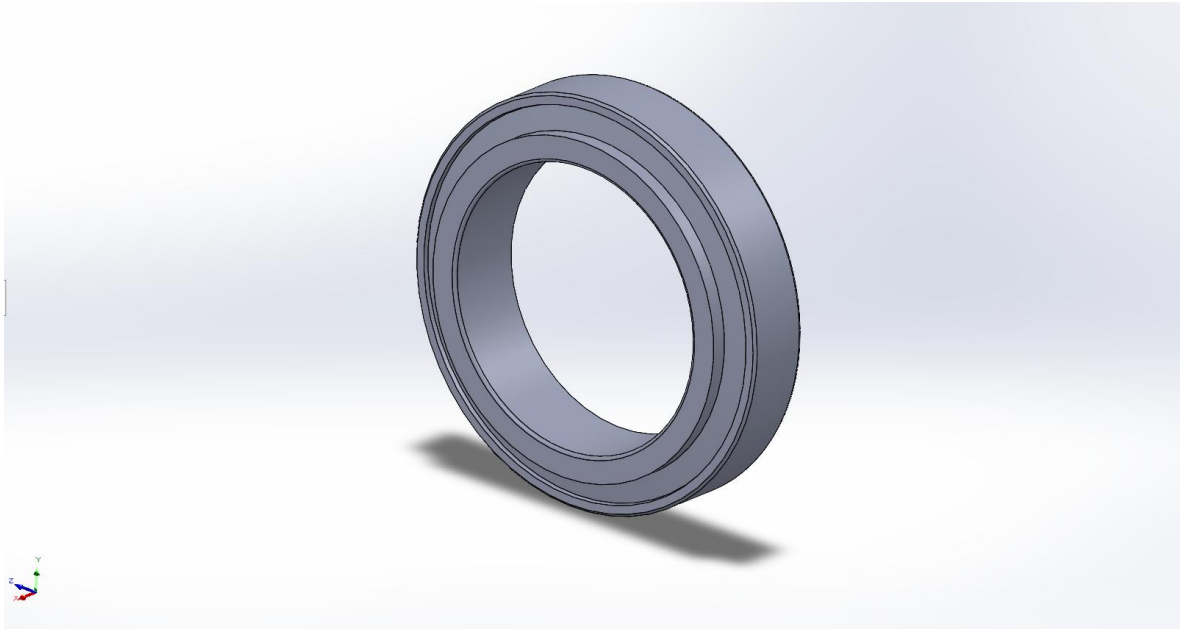


Figure 3.15 bearing

3.4. Simulation (ANSYS 16.2)

ANSYS is a powerful engineering simulation software that allows engineers to virtually analyze and predict the behavior of various physical systems. With modules like ANSYS Mechanical for structural analysis and ANSYS Fluent for fluid dynamics, it provides a comprehensive range of tools for simulating complex phenomena. Engineers can model and simulate designs, test different scenarios, and optimize performance before physical prototypes are built, reducing costs and time. ANSYS is widely used in industries such as aerospace, automotive, and manufacturing, enabling engineers to make informed design decisions and ensure product reliability and performance.

3.4.1. Workflow Diagram

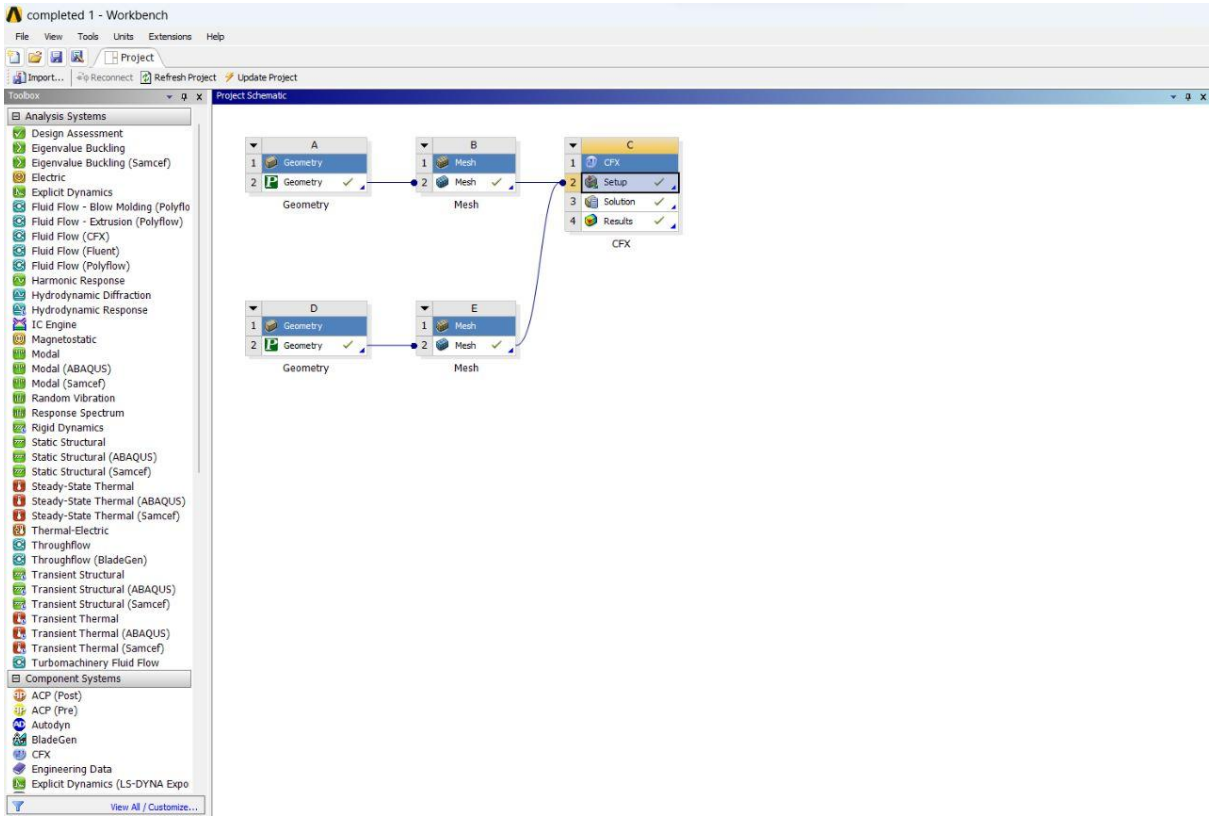


Figure 3.16 Workflow Diagram of simulation procedure

The workflow diagram for simulation in ANSYS consists of several interconnected steps that guide engineers through the process of conducting a simulation analysis. The initial step involves preparing the geometry of the system, either by creating a CAD model or importing an existing one. Subsequently, the geometry is discretized into small elements through meshing, ensuring appropriate density and quality. Material properties are assigned to the components, followed by defining boundary conditions and loading conditions. Solver setup involves specifying solution methods and convergence criteria. With the setup complete, the simulation is executed, and the solver calculates the system's response based on the defined conditions. Postprocessing tools are then utilized to visualize and analyze the results, enabling

engineers to extract relevant information such as stress distribution, fluid flow patterns, or temperature profiles. This data often guides engineers in making design modifications and optimizations. The simulation process in ANSYS is iterative, allowing engineers to refine the model and repeat the analysis until desired results and performance are achieved. Ultimately, the findings and conclusions are validated, documented, and communicated to stakeholders.

3.4.2. Geometry

Fluid domain of rotor and stator are made with solidworks 2016 and imported to ANSYS Workbench in Parasolid form respectively. To reduce computational effort we reduced the runner length and mass flow rate to one-fourth of the designed parameters. Because of the symmetricity the turbine efficiency would be similar.

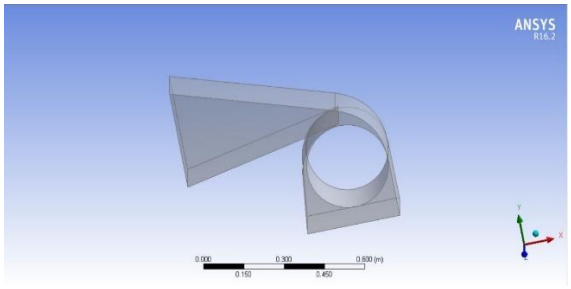


Figure 3.17 Stationary domain

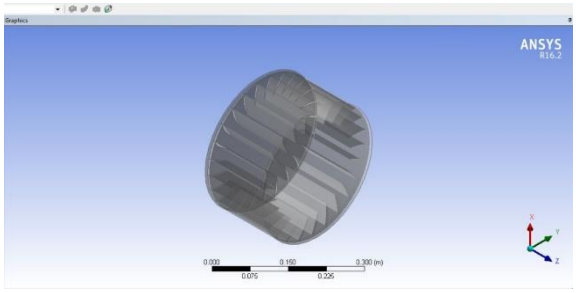


Figure 3.18 Rotary domain

3.4.3. Mesh

Meshing is done by ANSYS mesh in which rotating domain and stationary domain were meshed separately. The parameters for the meshing is given as:

Rotating domain

Physics Preference: CFD

Solver Preference: CFX

Element Type: Tetrahedrons

Inflation:

- Boundary layer thickness is calculated by Blasius solution with Reynolds number 150. Boundary layer thickness is calculated as 0.91mm. such that inflation layer of mesh cell with at least 5 times the thickness of local boundary layer is recommended. So that we used inflation option as First layer thickness with first layer height 0.1mm, maximum layer 5 and growth rate of 1.2.

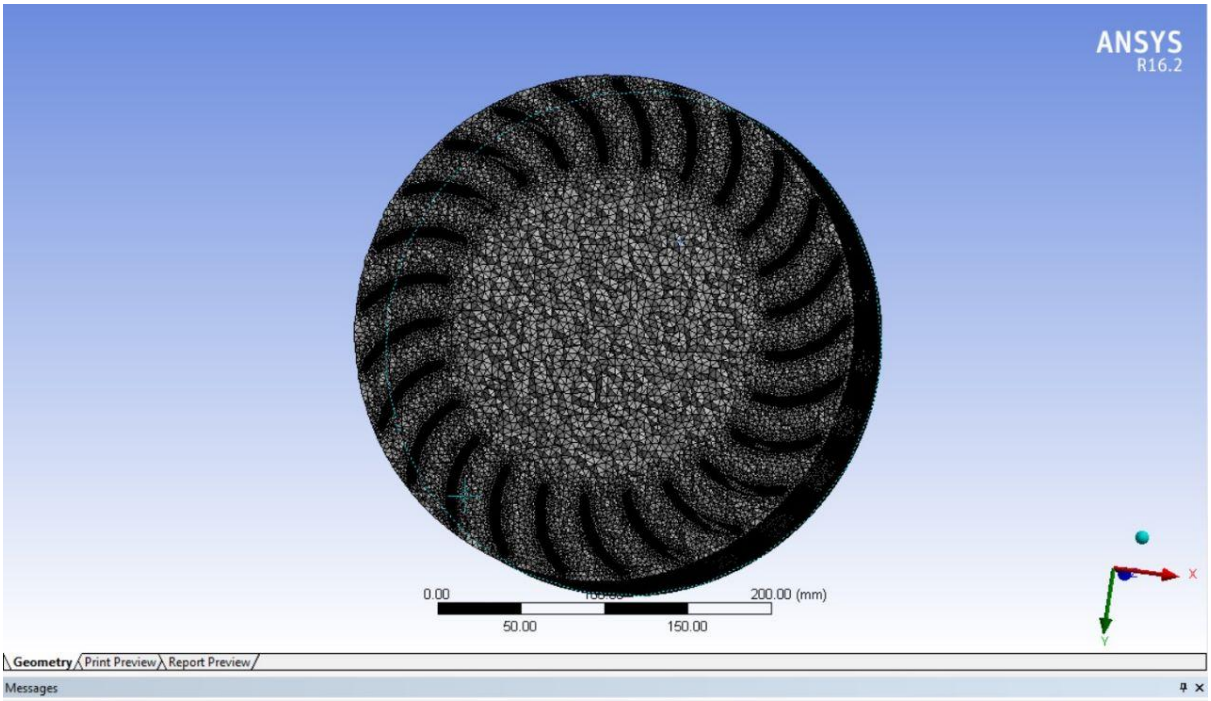


Figure 3.19 Rotating domain mesh

Stationary Domain:

Physics Preference: CFD

Solver Preference: CFX

Element Type: Tetrahedrons

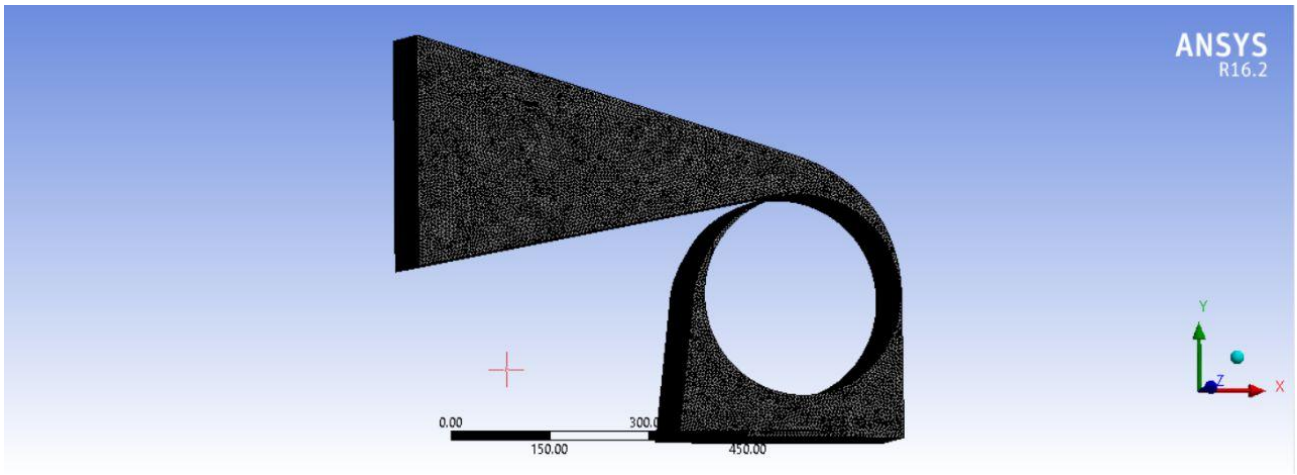


Figure 3.20 Stationary domain mesh

Mesh Statistics for Standard Blade domain

Nodes:145705

Elements:73861

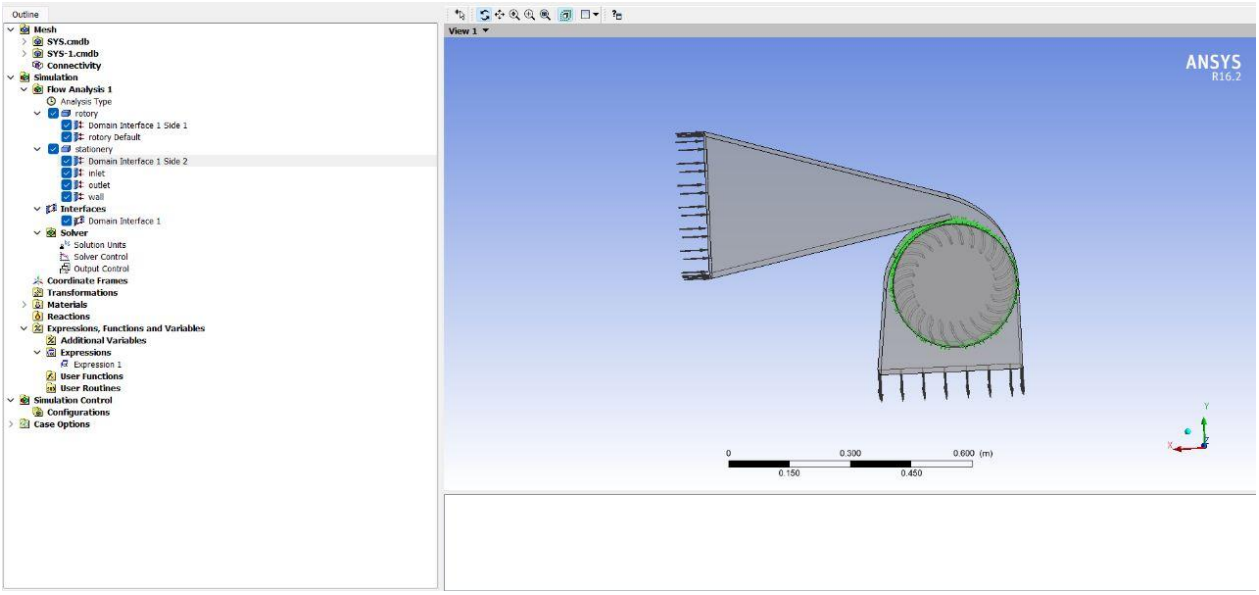
Mesh Statistics for Stationary domain

Nodes:68023

Elements:349841

3.4.4. CFX Pre

In CFX Pre the two meshed domain viz rotary and stationary are combined together. All the turbomachinery parameters are setup by using CFX tool 'Turbo Mode'. At the inlet the mass flow rate of 125 kg/s is applied whereas at the outlet average static pressure of 1atm is applied. Interface type of Frozen rotor is selected. k-epsilon turbulence model is selected. The k-epsilon turbulence model is commonly applied in ANSYS turbomachinery simulations due to its effectiveness in capturing turbulent flow behavior. Turbulence plays a significant role in turbomachinery applications, such as pumps, compressors, and turbines, where the flow is highly complex and unsteady.



3.4.5. Solution

In the CFX Solver the solution is processed in parallel environment with double precision and we use 7 partitions for parallel computations. The solution is carried out upto 100 iterations.

3.5. SIMULATION ANALYSIS

The simulation of turbine is done with two different kinds of blade profile. To find the efficiency of the turbine we simulated the models for the torque on rotor wall about Z-axis. The formula for finding efficiency of the turbine is

$$efficiency(\eta) = \frac{turbine\ power}{hydraulic\ power} \tag{9}$$

$$\text{Turbine power} = \frac{2\pi NT}{60000}\ \text{kw} \tag{10}$$

T= torque on rotor wall (Nm)

N= speed of turbine (RPM)

$$\text{Hydraulic power} = \rho QgH \tag{7}$$

ρ = density of water(1000 kg/m^3)

Q=volume flow rate of water

g = acceleration due to gravity(9.81 m/s^2)

H = net head available

3.5.1. Standard Blade Turbine

With the above methodology simulation on the standard blade turbine is done. The flow streamlines, velocity contour and pressure contour are obtained as given:

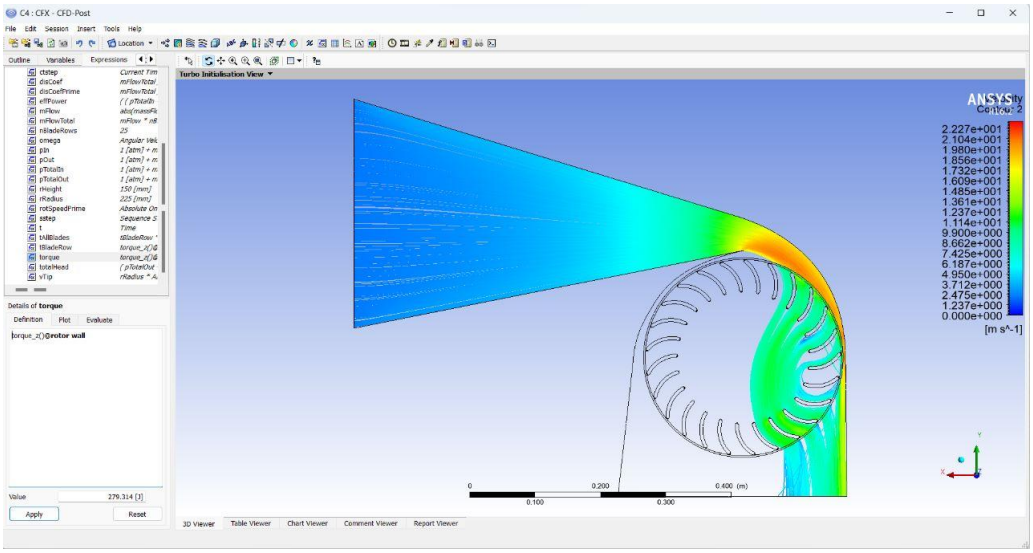


Figure 3.21 flow streamlines for standard blade turbine

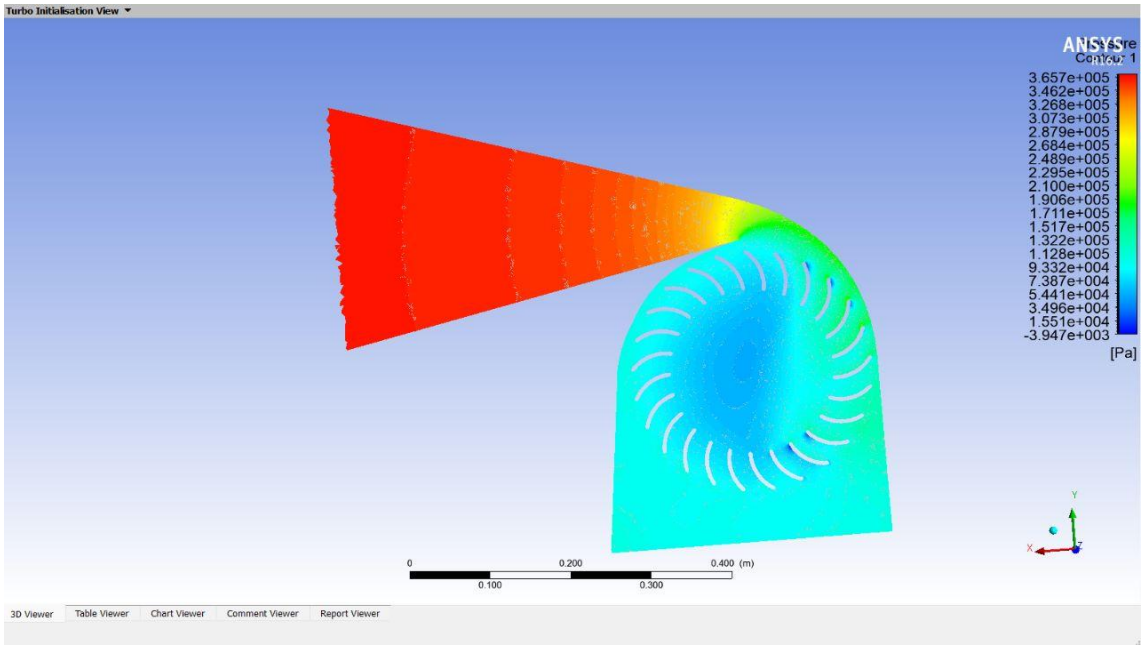


Figure 3.22 pressure contour for standard blade turbine

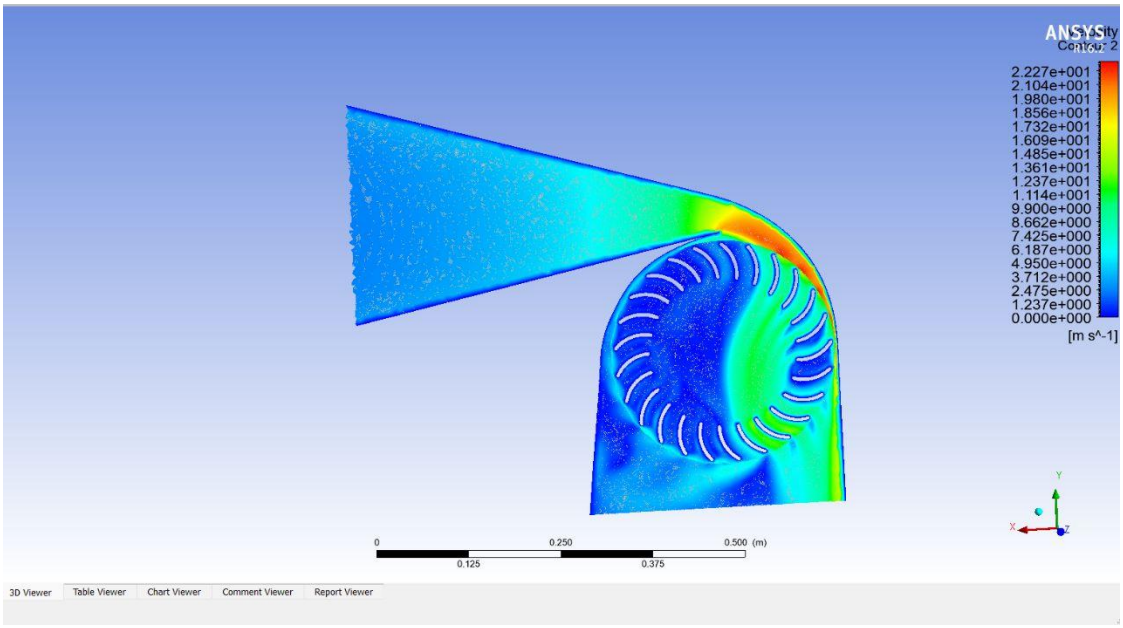


Figure 3.23 Velocity contour for standard blade turbine

The torque on the standard blade turbine is calculated by using the expression as

torque_z()@**rotor wall**

torque obtained = 279.314 Nm

efficiency = 71.6%

3.5.2. NACA 6409 blade turbine

For NACA airfoil blade turbine we use two different models with two kinds of blade profile inverted and airfoil orientation. Both orientation are as shown below

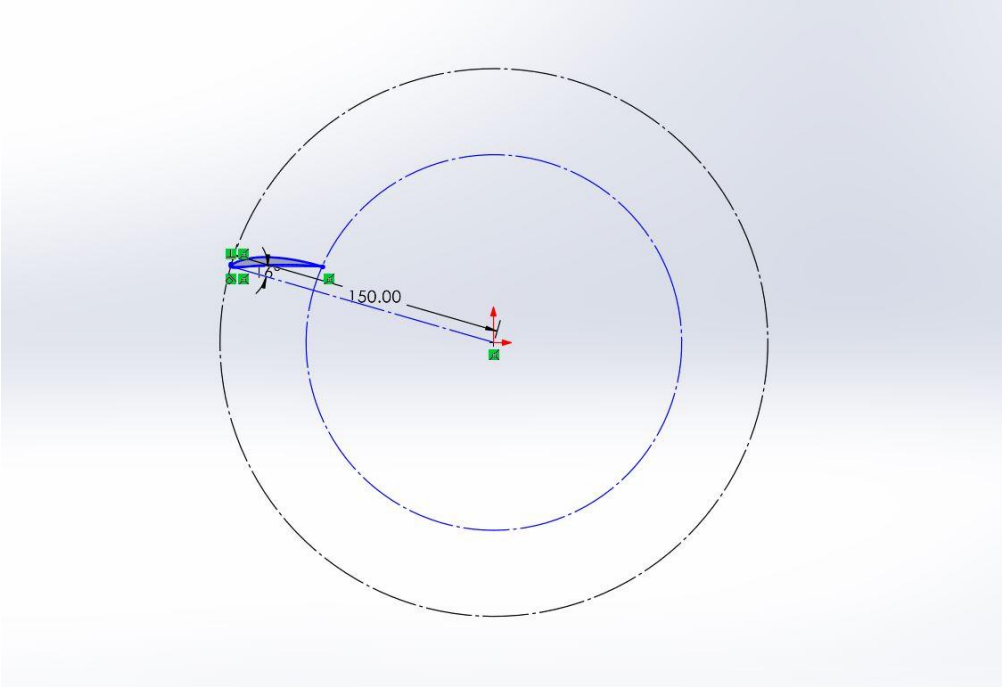


Figure 3.24 airfoil blade profile

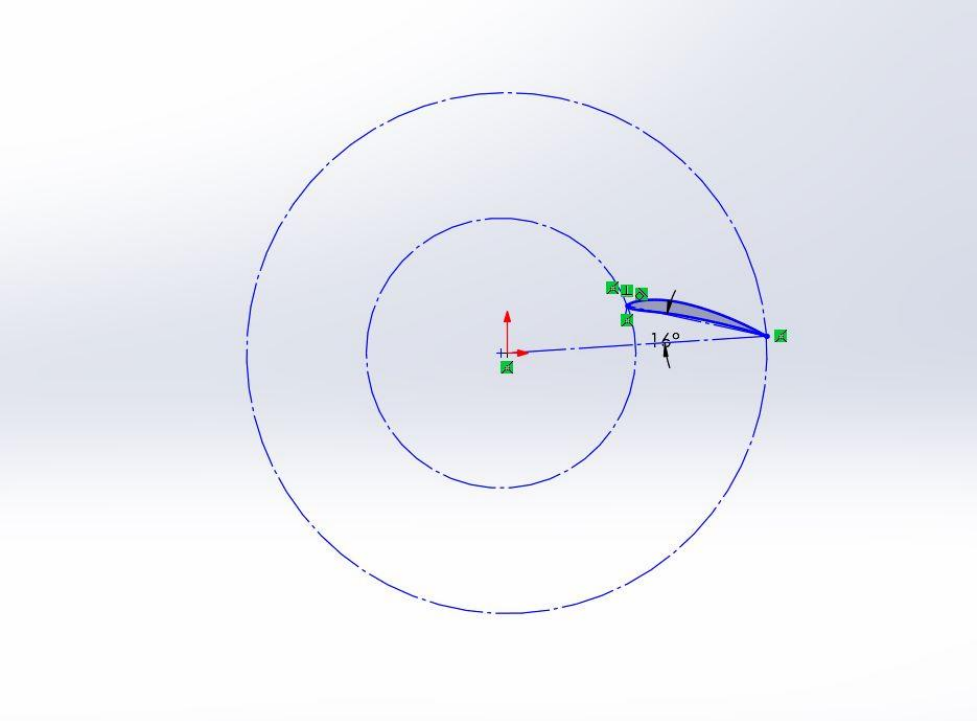


Figure 3.25 inverted blade profile

In the simulation of two blade profile shown above the airfoil blade profile gives more efficiency for the same chord length angle of 16^0. So, we simulated the blade profile for different chord length angle to find the optimum efficiency for the chord length angle. For airfoil blade turbine, the best efficiency is found at 29^0 chord length angle. The flow streamlines and pressure contour for the 29^0 chord length angle is shown as below

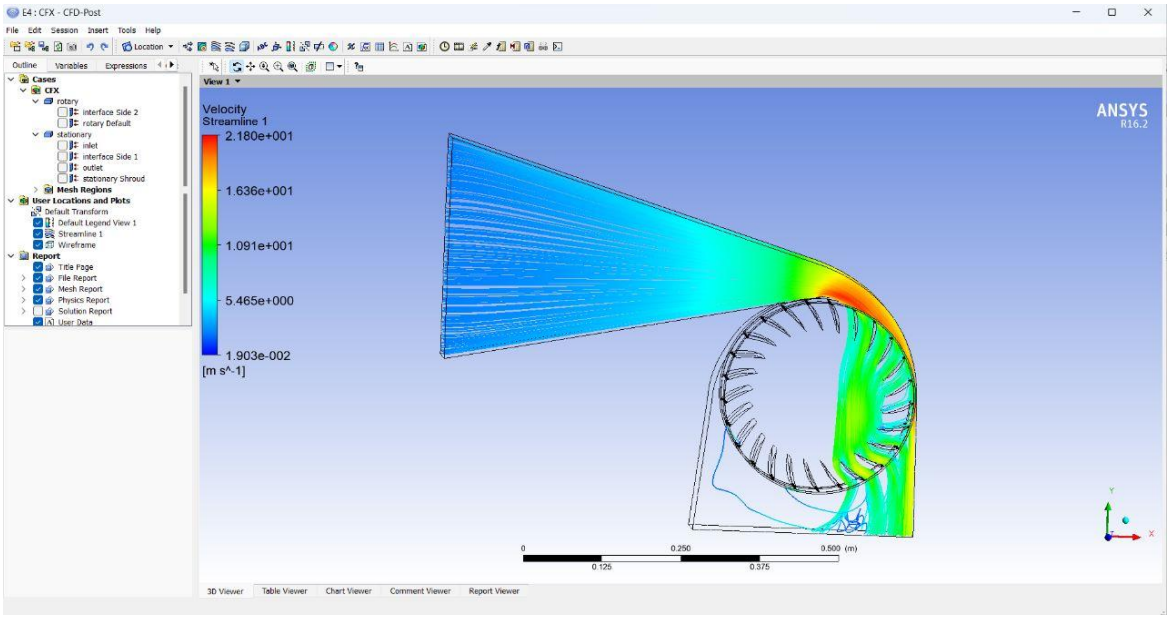


Figure 3.26 Flow streamlines for airfoil blade turbine

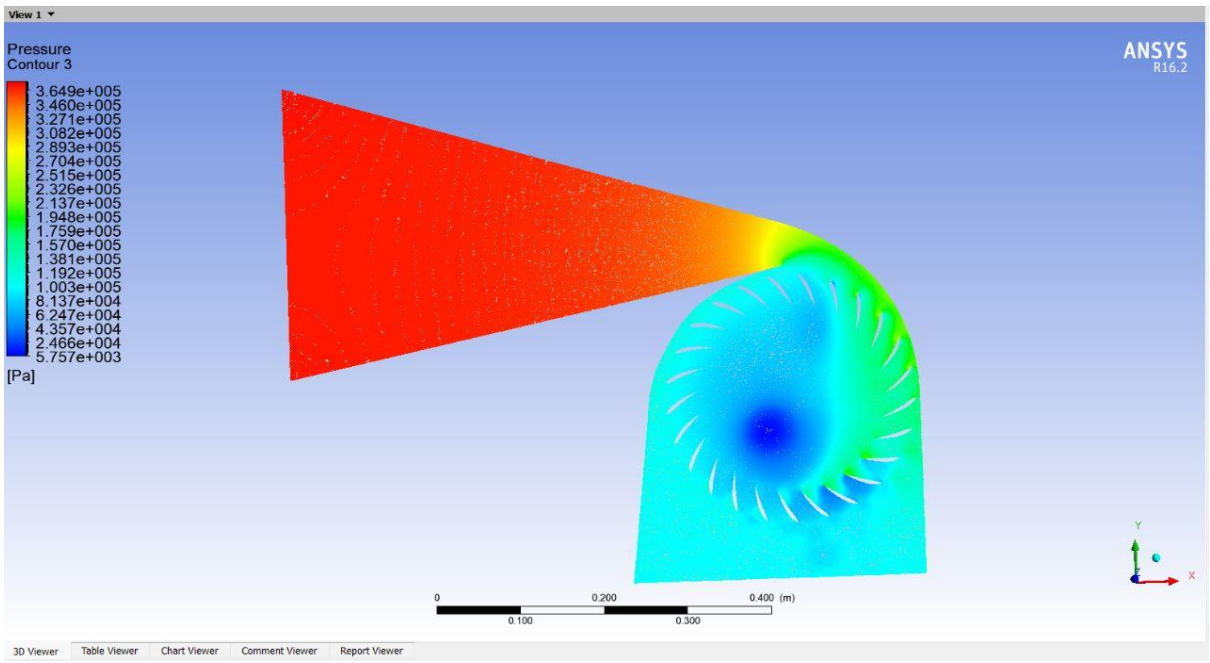


Figure 3.37 pressure contour for airfoil blade turbine

CHAPTER 4: RESULT

For inverted airfoil blade we simulated at 16^0 chord angle, the efficiency obtained at same chord angle for inverted blade is less than the airfoil blade. So, we only checked the efficiency of airfoil blade at varying chord angle. The efficiency of different blade configuration with varying chord angle of the turbine blade is as shown in the graph

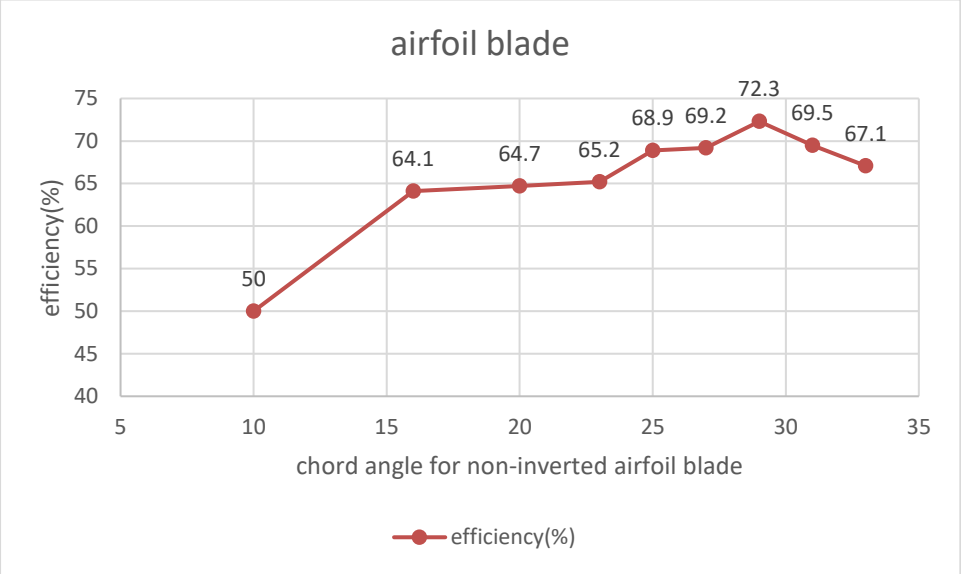


Figure 3.38 Efficiency vs Chord angle for airfoil blade

The above graph shows that the efficiency gradually increases from chord angle of 10^0 to 29^0. The maximum efficiency of 72.3% is obtained at chord angle of 29^0. After that the efficiency decreases with increasing chord angle. The efficiency of the standard blade turbine is found to be 70.6% while the airfoil airfoil blade with chord angle of 29^0 has 72.3% which is slightly greater than standard blade turbine.

CHAPTER 5: CONCLUSION AND RECOMENDATION

5.1. Conclusion

Thus, we have designed and simulate the standard blade turbine and airfoil blade turbines models in ANSYS and analyze all the results obtained and ended up on the conclusion that airfoil blade turbine which was designed has efficiency of 72.3% at 29^0 chord angle while the standard blade turbine has efficiency of 71.3% with specific design procedures and parameters obtained from the literature review.

Similarly we encountered the conclusion that the efficiency of the airfoil blade turbine increases to maximum 72.3% with increase on chord angle to maximum 29^0 and eventually efficiency of the airfoil blade turbine decreases after 29^0 chord angle.

5.2. Recommendations

- Comparison between efficiency and number of blades of turbine can be done by varying blade number.
- Turbine blade can be designed by turbo blade gen and meshing can be done in turbo mesh to get best quality mesh and complete turbo report for post processing if complete turbo report is required.

REFERENCES

1. W. Schleicher, "Characteristics of a micro-hydro turbine," Renewable and Sustainable Energy Reviews, no. 6, Jan 2014.
2. “A Year in Review. Nepal Electricity Authority,” Kathmandu, 2018.
3. “A year in review. Nepal electricity authority,” Kathmandu, 2020.
4. www.apec.gov.np, 28 may,2021.
5. N.koirala, R.Dhakal, D.Lubitiz, S.Bhandari “Reviews of Low Head Turbines system of Nepal for Rural Electrification”. 5-08-2017.
6. A.Laghari, “A comprehensive overview of new designs in the hydraulic, electrical equipment and controllers of mini hydro power plants making it cost effective technology,” Renewable and Sustainable energy reviews 20,pp 279-293,2013.
7. “Low Head Pico Hydro Promotion Project”,PEEDA,2009
8. Rajput R.K. ,”A Text Book of Fluid Machine”. 52-60 pp
9. Bilal A. N., “Design of High Efficiency Cross-Flow Turbine for Hydro-Power Plant”, International Journal of Engineering and Advanced Technology, ISSN: 2249 – 8958, Volume-2, Issue-3, 2013.
10. Krishna Nakarmi,”Harnessing Power on A Small Scale” volume-3
11. M. Sinagra et.al ,” Cross-Flow turbine design for variable operating conditions” Procedia Engineering 70 (2014) 1539 – 1548.
12. C.H. Achebe, et.al.” Design and implementation of a crossflow turbine for Pico hydropower electricity generation” (2020).
13. Manchan Tiwari , Rajendra Shrestha,” Effect of Variation of Design Parameters on Cross Flow Turbine Efficiency Using ANSYS”, Journal of the Institute of Engineering, 2017, 13(1): 1-9
14. CARIJA, Z, et.al,” FLUID FLOW SIMULATION OF A CROSSFLOW TURBINE”,Annals of DAAAM for 2009 & Proceedings of the 20th International DAAAM Symposium, Volume 20, No. 1, ISSN 1726-9679.

15. Dendy Adanta, et.al," Effect of Blade Depth on the Energy Conversion Process in Crossflow Turbines", CFD Letters 12, Issue 1 (2020) 123-131.
16. Young-Do Choi,et.al," Effect of Blade Angle on the Performance of a Cross-Flow Hydro Turbine", Journal of the Korean Society of Marine Engineering, Vol.32, No.3, pp.413-420,2008.
17. Myin San, Nyi Nyi," Design of Cross Flow Turbine and Analysis of Runner's Dimensions on Various Head and Flow Rate", International Journal of Scientific and Research Publications, Volume 8, Issue 8, August 2018

APPENDIX A: CALCULATION

Runner outer diameter

$D1 = \frac{40\sqrt{Hnet}}{rpm\ cross\ flow}$

$= \frac{40\sqrt{20}}{600}$

$= 0.298$

≈ 300 mm

Runner inner diameter

$\frac{D2}{D1} = 0.67$

D2=200mm

Jet Thickness

$So = 0.15 \times D1$

$= 0.045$

Length of Runner

$L = \frac{0.23Q}{So\sqrt{H}}$

$= 0.5714$m

≈ 600 mm

Design of cross flow turbine

Turbine expected output power

$= e_{\text{Turbine}} \times 10 \times Q \times H$

= 65kw

Range of specific speed of cross flow turbine

$40 \leq Ns \leq 200$

Applicable maximum specific speed

$N_s, max \leq 650\ H^{-0.5}$

Specific speed $N_s = \frac{513.55}{H^{0.505}}$

$= \frac{513.55}{20^{0.505}}$

$= 113.126$

$N = \frac{Ns\, h^{\frac{5}{4}}}{\sqrt{p}}$

$= \frac{113.126\ 20^{\frac{5}{4}}}{\sqrt{65}}$

$= 593.4$

600 rpm, $p_o = \frac{120f}{N}$

$= 10$

$K_{u1} = \frac{u1}{v1}$

$U_1 = \frac{\pi D_1 N}{60}$

$V_1 = k_c\sqrt{2gh}$, kc =0.98

$= 19.5$ m/s

$U_1 = 9$m/s

$D1 = \frac{60\times90}{\pi\times600}$

$= 0.286$

$=300$mm

BLADE PARAMETER

No of blades (Z) = 25

Blade thickness (t) = 5 mm

Radius of curvature of blades

r = 0.16×D1

r = 48 mm

Pitch circle radius

$R_o^2 = R_2^2 + r^2$

$R_o \approx 111$ mm

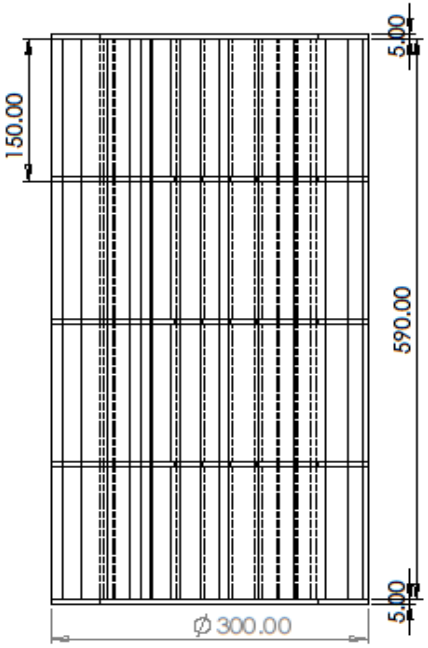
Predicted shaft diameter

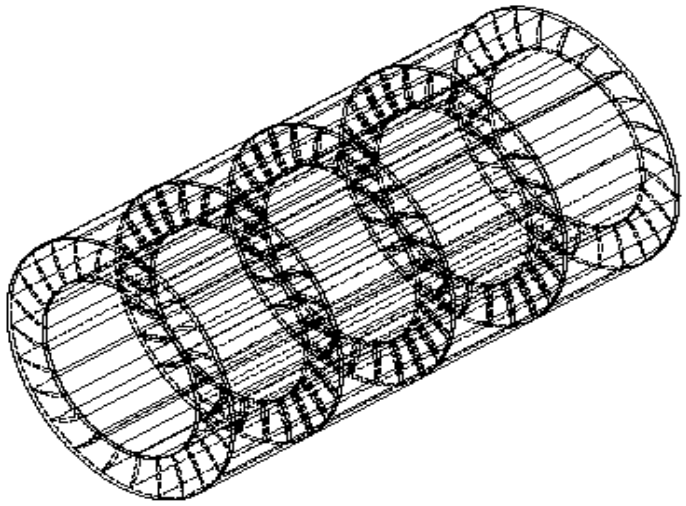
$D_s = 150 \sqrt[3]{\frac{p}{N}}$

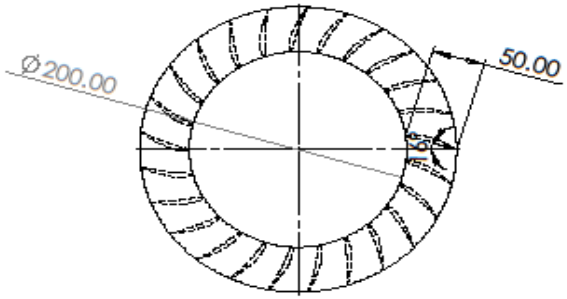
$= 150 \sqrt[3]{\frac{65}{600}}$

≈ 50mm

APPENDIX B: DRAWINGS

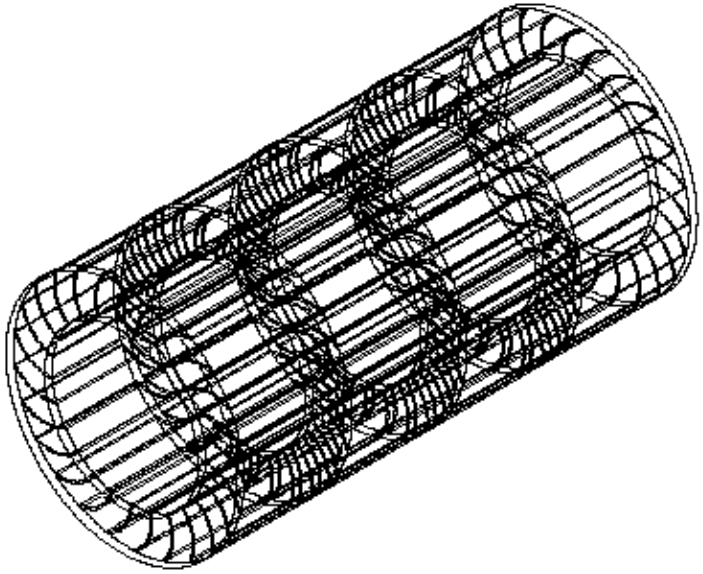


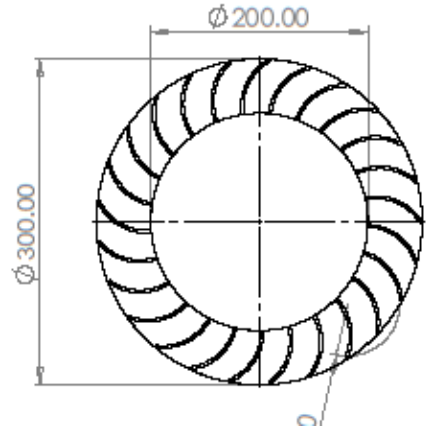




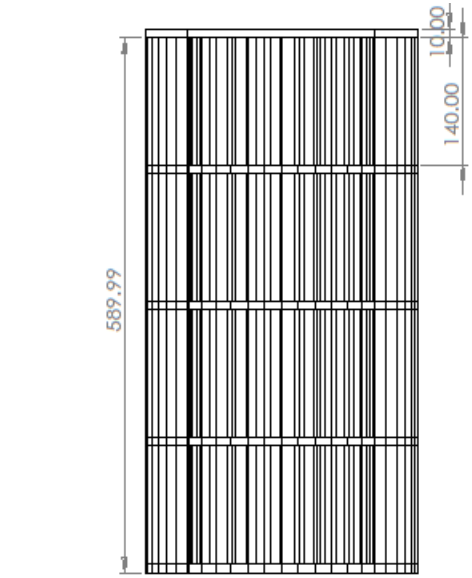
TITLE:	Inverted airfoil blade
SCALE 1:5	DWG NO.

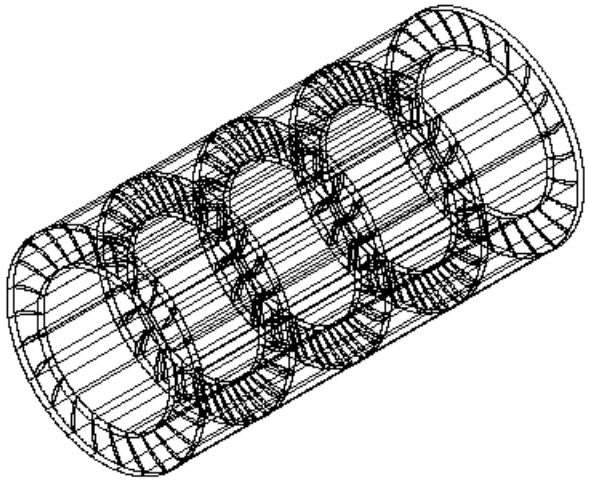


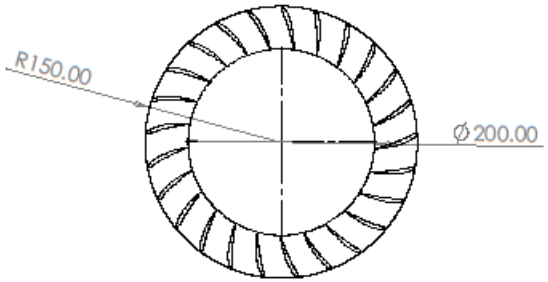




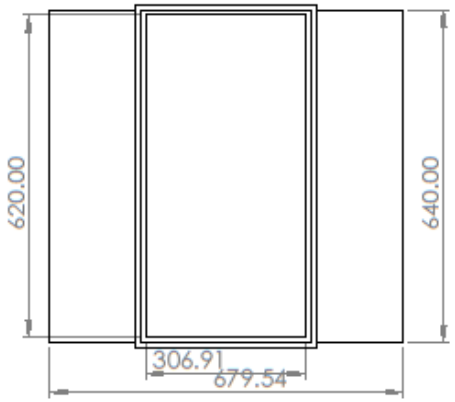
TITLE: standard blade	
SCALE	DWG NO.

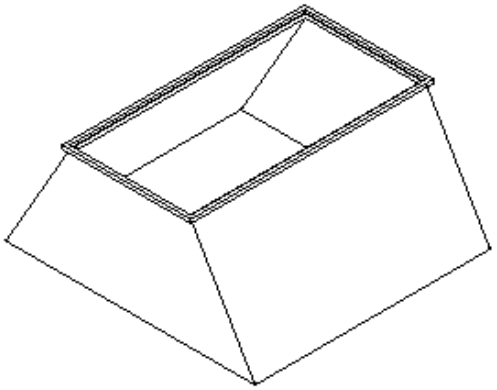


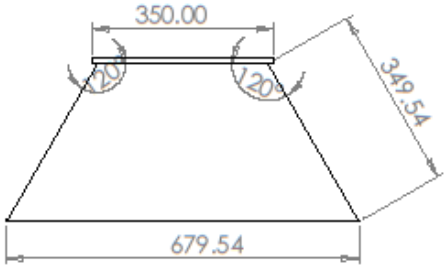




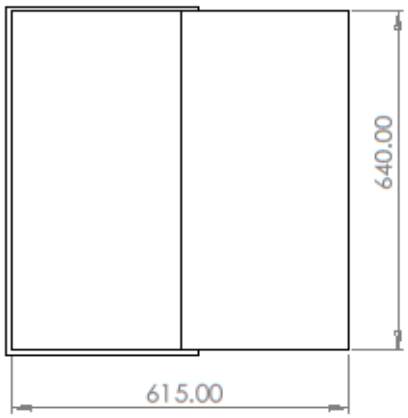
TITLE: Airfoil blade	
SCALE 1:5	DWG NO.

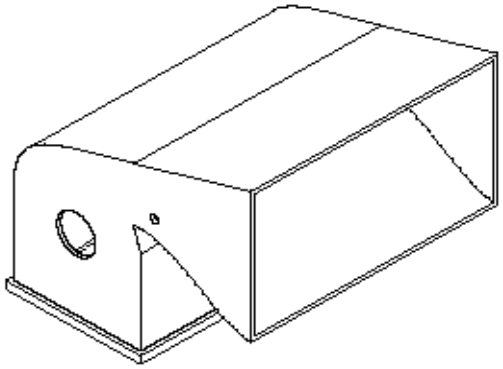


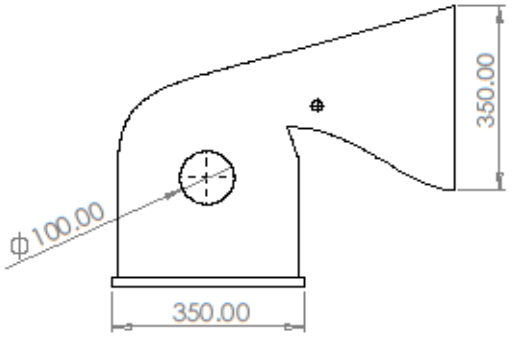




TITLE: Drafttube	
SCALE 1:10	DWG NO.







Title:	casing
scale: 1:20	DWG No.